Precautionary Retirement: Solvency Risk and the Claiming–Work Wedge

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Job Market Paper†

Latest version: Link

April, 2026

Abstract

We study how state-dependent risk of future Social Security benefit cuts changes the joint timing of claiming and retirement. When households attach a higher probability to a common future cut, they have an incentive to front-load protected payments by claiming earlier, but they also have a stronger incentive to self-insure by working longer and saving more. We develop a stylized mechanism and embed it in a life-cycle model disciplined by HRS moments on employment, claiming, wealth, health, and earnings, together with Trustees-based measures of fiscal strain and depletion risk.

Keywords: Social Security, retirement, claiming, policy uncertainty, trust-fund depletion.

JEL codes: H55, J14, J22, D84, E21.

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†I am deeply grateful to my advisor, Juan Carlos Cordoba, and my committee members, Rajesh Singh, Quinn Weninger, Sergio Lence, and Jian Li, for their thoughtful comments and sincere feedback.

1 Introduction

Social Security is the single largest source of retirement income in the United States, providing benefits to over 70 million Americans. For roughly half of beneficiaries aged 65 and older, it accounts for at least 50 percent of total income. The program's trust fund is projected to be depleted by the mid-2030s, at which point incoming payroll tax revenue would cover only 77 percent of scheduled benefits, triggering an automatic cut of roughly 23 percent. No act of Congress is required. Despite decades of Trustees Reports projecting this outcome, Congress has not legislated a policy reform since the 1983 Amendments. Therefore, how households who are approaching retirement respond to this growing fiscal risk is a first-order importance for both policy design and individual welfare.

The conventional approach treats claiming as a sufficient statistic for retirement: starting benefits and stopping work are the same decision. The Social Security Administration itself often uses the two terms interchangeably. Yet in the Health and Retirement Study (Figure 1), roughly 40 to 50 percent of claimers in birth cohorts 1935–1948 continue working after benefits begin. The two margins are not moving together: they are decoupling.

This paper argues that Social Security solvency-risk explains the wedge. When future benefits become less secure, early claiming becomes more attractive because it front-loads protected payments before a cut arrives. But the same risk also raises the value of continued work and private saving as a hedge against lower future income. A single solvency shock can therefore push claiming earlier and retirement later, opposite-direction responses on the two margins. To the best of our knowledge, this mechanism has not been studied in the retirement literature.

We embed this mechanism in a dynamic structural retirement model. The key innovation is a solvency-risk process tied to observable fiscal stress. Particularly, the probability of a common future benefit cut rises with the Social Security funding gap, and households internalize this risk when choosing when to claim, whether to work, and how much to save. We estimate the model by the method of simulated moments on HRS data, targeting moments on employment, wealth, and claiming across age, health, and permanent income. Prior structural work on retirement has focused on deterministic rule changes or cross-sectional incentive variation (Stock and Wise, 1990; French, 2005; French and Jones, 2011; Coile and Gruber, 2007; Coile et al., 2002; Benítez-Silva and Heiland, 2008; Gustman and Steinmeier, 2005). Instead, we depart by modeling how time-varying policy risk reshapes behavior before any reform is enacted.

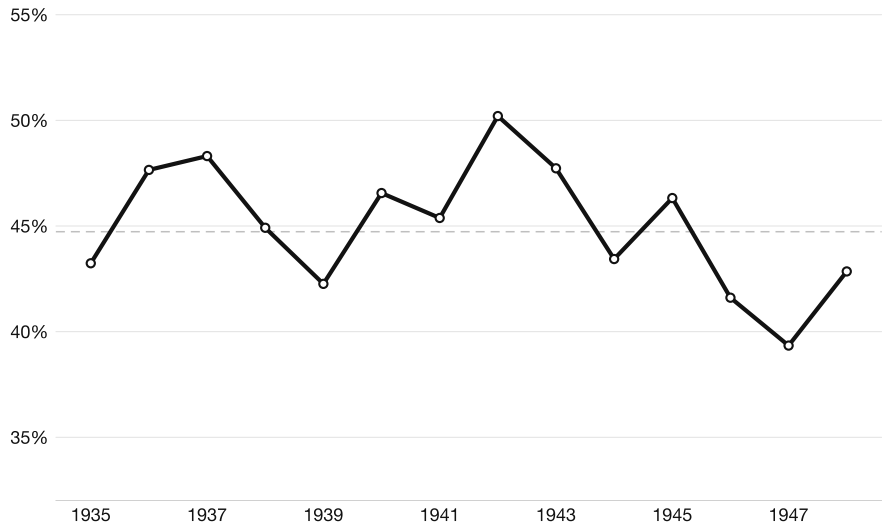

Figure 1: Life-cycle prevalence of work after claiming

Notes: The figure reports the weighted share of ever-claimers in birth cohorts 1935–1948 who work after claiming at least once between ages 60 and 75, using the HRS interview work-status measure. Across cohorts, the prevalence stays close to 2 in 5 to 1 in 2. The dashed line marks the cross-cohort weighted mean. This is a life-cycle prevalence object rather than a duration object. Source: Health and Retirement Study, 1992–2020.

Related Literature Our contribution sits at the intersection of three literatures. The first studies retirement incentives in dynamic settings with incomplete markets, health risk, and realistic Social Security rules (Rust and Phelan, 1997; French, 2005; French and Jones, 2011; Gustman and Steinmeier, 2005). We add an observable solvency-risk process that generates time-varying policy uncertainty over the life cycle. The second emphasizes that claiming and retirement are separate margins, both empirically and structurally (Coile et al., 2002; Benítez-Silva and Heiland, 2008; Pashchenko and Porapakkarm, 2024; Bairoliya and McKiernan, 2026). We build on these contributions by showing how a common source of policy risk moves both margins simultaneously and in opposite directions. The third studies information, beliefs, and policy uncertainty in Social Security (Liebman and Luttmer, 2015; Luttmer and Samwick, 2018; Shoven et al., 2017). We connect observed fiscal stress to state-dependent cut probabilities and trace the behavioral consequences for claiming, work, and saving before any reform is implemented.

Roadmap The rest of the paper is organized as follows. Section 2 develops a stylized mechanism that isolates the joint claiming and self-insurance margins. Section 3 embeds that mechanism in a quantitative life-cycle model. Section 4 describes the data and measurement. Section 5 presents the calibration and estimation. Section 6 reports model fit and main quantitative

results. Section 7 presents policy counterfactuals. Section 8 discusses robustness and external validation. Section 9 concludes.

2 Stylized Mechanism

A rise in solvency risk changes two margins at once. When future benefits become less secure, early claiming becomes more attractive because it front-loads protected payments. The same increase in risk also raises the value of private buffers, so continued work and saving become more attractive. This section isolates that mechanism in the simplest possible environment. The stylized model focuses on one reduced-form object, the probability p that a common future benefit cut applies to promised Social Security payments.

2.1 Environment

Time is discrete. In every period a unit mass of agents is born, and each cohort lives for three periods: a pre-retirement period $j = 0$, an early-retirement period $j = 1$, and an old-age period $j = 2$. We study one cohort's problem in partial equilibrium, taking the interest rate r and wages w_j as given.

Agents can work in periods $j = 0$ and $j = 1$. Labor supply is binary, $n_j \in \{0, 1\}$, and old-age labor supply is zero. Preferences are

$$U = u(c_0) - \psi_0 n_0 + \beta\big[u(c_1) - \psi_1 n_1\big] + \beta^2 u(c_2),$$

where $u' > 0$, $u'' < 0$, and $\psi_j > 0$ is the utility cost of work. Assets are risk-free and cannot be negative.

The Social Security timing problem is deliberately simple. In period $j = 1$, the household either claims immediately and receives the reduced benefit $b_E > 0$ in periods 1 and 2, or waits until period 2 and receives the larger delayed-claim benefit $b_D > b_E$ only in period 2. Claiming is irreversible. The key state at the retirement-claiming date is (a_1, p), where a_1 is beginning-of-period assets and p is the probability that a common benefit cut applies between periods 1 and 2.

2.2 Solvency Risk

The stylized model summarizes fiscal conditions by an observed scalar G that captures Social Security fiscal stress. Higher fiscal stress makes a future common cut more likely:

$$p = h(G), \qquad h'(G) > 0,$$

with $p \in [0, 1]$. We do not model a separate latent fiscal state in the main text. Instead, p is the reduced-form summary of the policy environment that households face when they choose whether to work and whether to claim.

Between periods 1 and 2, a common cut may occur. Let

$$\omega \in \{0, 1\},$$

where $\omega = 0$ denotes the promised regime and $\omega = 1$ denotes the cut regime. If $\omega = 1$, all period-2 Social Security payments are scaled down by the factor $(1 - \delta)$, with $\delta \in (0, 1)$. Let $d \in \{E, D\}$ denote the claim decision. The promised period-2 benefit is b_d, and actual period-2 benefits are

$$(1 - \delta\omega)b_d.$$

Benefits already received in period 1 are protected. This asymmetry is the source of the claiming margin: early claiming shortens the effective duration of the household's exposure to the public pension, at the cost of a smaller benefit base.

2.3 The Household Problem

We solve the household's problem by backward induction.

Period 2. Given assets a_2 carried into old age, old-age consumption is

$$c_2 = (1 + r)a_2 + (1 - \delta\omega)b_d,$$

where $d \in \{E, D\}$ is the earlier claim decision and $\omega \in \{0, 1\}$ is the realized policy regime.

Define the expected old-age continuation value

$$\begin{aligned} \mathcal{J}(a_2, p; b) &\equiv \mathbb{E}_\omega\big[u\big((1 + r)a_2 + (1 - \delta\omega)b\big) \mid p\big] \\ &= (1 - p)\, u\big((1 + r)a_2 + b\big) + p\, u\big((1 + r)a_2 + (1 - \delta)b\big). \end{aligned} \tag{1}$$

This object summarizes the value of entering old age with assets a_2 when future benefits of size b are exposed to cut risk p.

Period 1. Conditional on claim choice $d \in \{E, D\}$, the period-1 problem is

$$V_1^d(a_1, p) = \max_{n_1 \in \{0,1\},\, a_2 \geq 0} \{u(c_1) - \psi_1 n_1 + \beta \mathcal{J}(a_2, p; b_d)\}, \tag{2}$$

subject to

$$c_1 + a_2 = (1+r)a_1 + w_1 n_1 + b_E \mathbf{1}\{d = E\}.$$

Thus $d = E$ gives the current-period benefit flow b_E, while $d = D$ gives no period-1 benefit and a larger promised benefit base b_D in period 2. Define

$$V_1(a_1, p) \equiv \max\{V_1^E(a_1, p),\, V_1^D(a_1, p)\}.$$

Period 0. Before the claiming decision arrives, the household chooses work and saving in anticipation of next period's policy environment:

$$V_0 = \max_{n_0 \in \{0,1\},\, a_1 \geq 0} \{u(c_0) - \psi_0 n_0 + \beta \mathbb{E}_G [V_1(a_1, h(G))]\}, \tag{3}$$

subject to

$$c_0 + a_1 = w_0 n_0.$$

Households therefore choose period-0 saving in anticipation of the distribution of fiscal stress they may face at the retirement-claiming date.

2.4 Understanding the Mechanism

The stylized model delivers two margins that move in opposite directions when solvency risk rises.

2.4.1 Work and Saving margin

Because the utility function u is concave and the cut regime delivers lower old-age consumption than the promised regime, the marginal value of old-age assets is higher when policy risk is higher:

$$\frac{\partial^2 \mathcal{J}(a_2, p; b)}{\partial a_2 \partial p} = (1+r)\left[u'\big((1+r)a_2 + (1-\delta)b\big) - u'\big((1+r)a_2 + b\big)\right] > 0.$$

Policy risk therefore strengthens the precautionary motive to work and save.

To make this explicit, let $\mathcal{W}_1(a_1, p; n_1)$ denote the period-1 value when the work choice is fixed at $n_1 \in \{0, 1\}$ and the household optimizes over claiming and saving:

$$\mathcal{W}_1(a_1, p; n_1) = \max_{d \in \{E,D\},\, a_2 \geq 0} \Big\{ u\Big((1+r)a_1 + w_1 n_1 + \mathbf{1}\{d = E\} b_E - a_2\Big) - \psi_1 n_1 + \beta \mathcal{J}(a_2, p; b_d) \Big\}.$$

Define the work surplus

$$\Delta^W(a_1, p) \equiv \mathcal{W}_1(a_1, p; 1) - \mathcal{W}_1(a_1, p; 0).$$

To sign $\partial \Delta^W / \partial p$, it is useful to introduce one object that will also reappear on the claiming margin. Let

$$L(a_2, b) \equiv u\big((1+r)a_2 + b\big) - u\big((1+r)a_2 + (1-\delta)b\big) \geq 0$$

denote the utility loss from a benefit cut when the household enters old age with assets a_2 and promised benefit b. By the envelope theorem applied to $\mathcal{W}_1$,

$$\frac{\partial \mathcal{W}_1(a_1, p; n_1)}{\partial p} = -\beta L\big(a_2^*(n_1),\, b_{d^*(n_1)}\big),$$

where $(d^*(n_1), a_2^*(n_1))$ are the optimal claiming and saving choices conditional on n_1. Signing $\partial \Delta^W / \partial p$ therefore reduces to comparing L at the work optimum and at the no-work optimum. Two observations deliver the sign. First, working adds $w_1 > 0$ to period-1 cash-on-hand, so by the standard consumption–saving argument $a_2^*(1) > a_2^*(0)$: labor income is partly converted into the private asset position that the household carries into old age. Second, differentiating L with respect to a_2 gives

$$\frac{\partial L(a_2, b)}{\partial a_2} \;=\; (1+r)\Big[u'\big((1+r)a_2 + b\big) - u'\big((1+r)a_2 + (1-\delta)b\big)\Big] \;<\; 0,$$

because u is concave and $(1+r)a_2 + b > (1+r)a_2 + (1-\delta)b$. A larger private buffer narrows the marginal-utility gap between the promised and the cut regime, so a given cut is less painful when assets are higher. Combining the two observations, the work optimum has both higher a_2 and, therefore, a strictly smaller realized L than the no-work optimum, so

$$\frac{\partial \Delta^W(a_1, p)}{\partial p} = -\beta\Big[L\big(a_2^*(1), \cdot\big) - L\big(a_2^*(0), \cdot\big)\Big] > 0.$$

The economic interpretation is that labor income is a hedge against the cut. Unlike the future

Social Security benefit, wages are not exposed to the policy risk and can be converted into private assets that smooth consumption across the two regimes. The higher is p, the more valuable this hedge becomes, and the stronger is the incentive to supply labor in period 1.

Note that the sign of $\partial\Delta^W/\partial p$ does not depend on the claim choice. If working and not working lead to the same claim, the comparison of L is a pure asset-channel move along a_2. If working tilts the claim, the envelope theorem still delivers the result: by the cross-partial $\partial^2\mathcal{J}/\partial a_2\partial p > 0$ established above, extra private wealth is more valuable when p is higher under either claim branch, so the work surplus rises in p regardless of which branch is optimal. The work margin therefore responds to solvency risk through the asset channel alone, independent of the claiming decision.

2.4.2 Claiming margin

The same object $L(a_2, b)$ also governs the claiming margin. Define the claim-now surplus

$$\Delta^C(a_1, p) \equiv V_1^E(a_1, p) - V_1^D(a_1, p).$$

Applying the envelope theorem separately to each claim branch, and noting that the only probability p dependent object in V_1^d is the continuation value $\mathcal{J}$,

$$\frac{\partial V_1^d(a_1, p)}{\partial p} = \beta\frac{\partial\mathcal{J}\big(a_2^*(d), p; b_d\big)}{\partial p} = -\beta L\big(a_2^*(d),\, b_d\big),$$

where $a_2^*(d)$ is the optimal saving choice conditional on claim d. Subtracting,

$$\frac{\partial\Delta^C(a_1, p)}{\partial p} = \beta\Big[L\big(a_2^*(D),\, b_D\big) - L\big(a_2^*(E),\, b_E\big)\Big].$$

The claiming margin therefore comes down to whether the realized cut loss L is larger at the delayed-claim optimum than at the early-claim optimum. Two independent channels, both pushing in the same direction, deliver this.

The first is an asset channel. Early claiming adds b_E to period-1 cash-on-hand while delayed claiming does not, so by the same consumption–saving argument used on the work margin, $a_2^*(E) > a_2^*(D)$. Early claiming pulls a portion of the public annuity forward into period 1, which is immune to any cut realized between periods 1 and 2, and a share of that cash is saved, increasing the private buffer that enters old age. Because $\partial L/\partial a_2 < 0$ under concavity of u,

the larger buffer reduces the realized cut loss, holding benefit b_E fixed,

$$L\big(a_2^*(E),\, b_E\big) < L\big(a_2^*(D),\, b_E\big).$$

The second is a benefit-base channel. Holding a_2 fixed, differentiate L with respect to b:

$$\frac{\partial L(a_2, b)}{\partial b} \;=\; u'\big((1+r)a_2 + b\big) \;-\; (1-\delta)\, u'\big((1+r)a_2 + (1-\delta)b\big).$$

This is positive under standard curvature: for log and CRRA utility with $\sigma \leq 1$ the expression is positive for all b, and for $\sigma > 1$ it is positive whenever the private buffer $(1+r)a_2$ is not negligible relative to the promised benefit b, which is the empirically relevant region for near-retirement households whose median private wealth is several times annual Social Security benefits. A larger promised base therefore implies a larger absolute dollar loss from a proportional cut. Because $b_D > b_E$,

$$L\big(a_2^*(D),\, b_D\big) > L\big(a_2^*(D),\, b_E\big).$$

Combining the two inequalities,

$$L\big(a_2^*(E),\, b_E\big) < L\big(a_2^*(D),\, b_E\big) < L\big(a_2^*(D),\, b_D\big),$$

so $\partial \Delta^C(a_1, p)/\partial p > 0$. That is a rise in solvency risk makes immediate claiming more attractive. The economic interpretation is that delayed claiming has both less protective private wealth and a larger absolute cut to absorb, while early claiming shortens the duration of exposure to the solvency risk by pulling a protected payment into period 1.

Claiming as maturity choice. The claiming margin has a natural portfolio interpretation. Early claiming converts deferred, politically exposed pension promises into a shorter-duration annuity stream that begins immediately. Delayed claiming raises the benefit base but extends the duration of exposure to legislative risk. The comparative static $\partial \Delta^C/\partial p > 0$ is therefore a standard substitution result: when the long-duration public asset becomes riskier, the household shifts toward the shorter-duration alternative and toward private safe assets that are not subject to political haircut (Shoven and Slavov, 2014). Together with the precautionary work-and-saving margin, this implies that a single policy-risk object generates portfolio rebalancing on both margins simultaneously.

Distributional implication. The same mechanism yields a clear heterogeneity prediction. Households with fewer private assets are less able to smooth a future benefit cut through self-insurance. They therefore sit closer to both the work cutoff and the claim-now cutoff, so an increase in p should move lower-asset households more strongly on both margins. Equivalently, lower-asset households hold a larger share of lifetime resources in the politically exposed public annuity, so the portfolio rebalancing triggered by higher p is proportionally larger.

2.5 Connecting to the Quantitative Model

The theory isolates two channels. The first is a *precautionary work-and-saving* channel: when future benefits become less secure, the marginal value of private assets rises, which increases the value of working and saving. The second is a *claiming front-loading* channel: when future benefits are at risk, the option value of waiting falls and the current-period payment from early claiming becomes more attractive. Together, these channels decouple claiming from retirement.

Section 3 preserves this mechanism while replacing the reduced-form state p with an observed aggregate fiscal process. In the quantitative model, fiscal stress evolves through the funding gap G_t, and the probability of entering the common cut regime $\mu_{t+1} = 1$ is an increasing function of updated fiscal stress G_{t+1}. The quantitative model therefore keeps the same economic logic as the stylized theory while tying policy risk to observables used in calibration.

3 Quantitative Model

Consider an older household seeking to maximize expected discounted lifetime utility from age 55 onward. The quantitative model is an annual partial-equilibrium life-cycle framework in which households choose whether to work, how much to save, and when to claim Social Security, subject to health risk, survival risk, realistic benefit rules, and the possibility of a common future benefit cut. Following French (2005) and French and Jones (2011), the household faces a discrete extensive labor-supply margin and derives warm-glow utility from bequests. The key departure from standard retirement models is a policy-risk block: an observed measure of fiscal stress governs the probability that a common benefit reduction arrives before the household dies, and the household responds by jointly adjusting claiming, work, and saving.

3.1 Environment

Time is indexed by age $j \in \{55, \ldots, J\}$, where $J = 90$. Each household belongs to a birth cohort c, so that calendar year is $t = c + j$. The solvency-risk index G_t is an observed aggregate object

that varies across calendar years but not across households within the same year. Because G_t is deterministic and publicly observed, it is not part of the household's endogenous state; instead, it enters the problem as a cohort-specific exogenous input. The model is therefore solved separately for each cohort, with each cohort facing the realized path $\{G_{c+55}, G_{c+56}, \ldots, G_{c+J}\}$.

At the beginning of age j, household i in cohort c enters the period with state

$$\mathbf{x}_{ij} = (a_{ij}, \bar{A}_{ij}, q_{ij}, \omega_j, \theta_i, \nu_i, H_{ij}),$$

where a_{ij} denotes beginning-of-period assets, $\bar{A}_{ij}$ is the household's Social Security earnings state measured in monthly AIME units, q_{ij} records claiming status and age, $\omega_j \in \{0, 1\}$ is the aggregate policy regime, $\theta_i \in \{1, \ldots, 5\}$ is a permanent earnings type, $\nu_i \in \{1, \ldots, N_\gamma\}$ is a persistent preference type, and $H_{ij} \in \{1, 2, 3\}$ is health.

Assets a_{ij} summarize self-insurance and intended bequests. The earnings state $\bar{A}_{ij}$ is the sufficient statistic for future retired-worker benefits. The claiming indicator satisfies $q_{ij} = 0$ before benefits begin and $q_{ij} \in \{62, \ldots, 70\}$ after claim. The aggregate regime $\omega_j = 0$ denotes the promised regime and $\omega_j = 1$ denotes the cut regime; once realized, the cut is permanent. Permanent type θ_i captures fixed lifetime earnings differences. Preference type ν_i captures persistent heterogeneity in the consumption-leisure tradeoff. Health H_{ij} shifts both survival and the time cost of work.

The household chooses whether to work, how much to save, and — when eligible — whether to initiate Social Security benefits. The choice set is therefore

$$n_{ij} \in \{0, 1\}, \qquad a_{i,j+1} \geq 0, \qquad \iota_{ij} \in \{0, 1 - \mathbf{1}\{q_{ij} > 0\}\},$$

where $\iota_{ij} = 1$ denotes a new claim initiation and is available only to unclaimed households between ages 62 and 70. Before age 62, $\iota_{ij} = 0$. At age 70, any remaining unclaimed household is forced to claim. There is no intensive labor-supply choice. By abstracting from hours variation, the model concentrates the late-life labor-supply response on the participation margin, which is the dominant margin of adjustment among older workers (French, 2005). Consumption is determined residually from the budget constraint.

Initial Conditions. Households enter the model at age 55. The joint distribution of permanent earnings type and health at entry is taken from the data. Conditional on permanent earnings type, initial assets and initial AIME are drawn from type-specific distributions estimated outside the model. Preference heterogeneity is layered on top of that empirical entry distribution:

conditional on earnings type, households are assigned to one of the persistent preference types described below. All households enter unclaimed ($q_{55} = 0$) and in the promised regime ($\omega_{55} = 0$).

Timing Within the Period. Within each age j, the timing is as follows. First, the household observes its state $\mathbf{x}_{ij}$, including the current policy regime. Second, if the household is eligible and chooses to claim, benefits begin immediately. Third, the household chooses work and saving. Consumption is determined residually, and end-of-period assets are carried forward. Finally, health evolves, survival is realized, and the policy regime may transition. If death occurs, the household receives warm-glow utility from bequeathed assets.

3.2 Preferences

Following French (2005) and French and Jones (2011), the household values both consumption and leisure. Period utility is

$$u(c_{ij}, \ell_{ij}, n_{ij}) = \frac{\left(c_{ij}^{\gamma_{\nu_i}} \ell_{ij}^{1-\gamma_{\nu_i}}\right)^{1-\sigma} - 1}{1-\sigma} - \psi(H_{ij})\, n_{ij},$$

with the usual logarithmic limit when $\sigma = 1$. The parameter $\sigma > 0$ governs curvature over the consumption-leisure composite. The function $\psi(H) \geq 0$ is the utility cost of working, which depends on health status. We parameterize it as $\psi(1) = \psi_0$, $\psi(2) = \psi_0 + \frac{1}{2}\delta$, and $\psi(3) = \psi_0 + \delta$, where $\psi_0 \geq 0$ is the baseline work cost and a single parameter $\delta \geq 0$ governs the health gradient with a fixed fair-to-poor ratio. The parameter $\gamma_{\nu_i} \in (0, 1)$ determines the household's relative taste for consumption versus leisure. Higher values of γ_{ν_i} place more weight on consumption and therefore weaken the leisure motive holding everything else fixed.

We allow for persistent heterogeneity in this consumption weight. At age 55, each household draws a type ν_i that remains fixed over the life cycle. When $N_\gamma = 3$, the distribution of γ_{ν_i} is approximated by the three-point support

$$\gamma_\nu \in \left\{\gamma - \sqrt{3}\sigma_\gamma,\ \gamma,\ \gamma + \sqrt{3}\sigma_\gamma\right\}, \qquad \Pr(\nu = 1, 2, 3) = \left(\tfrac{1}{6}, \tfrac{2}{3}, \tfrac{1}{6}\right),$$

so that the mean is γ and the standard deviation is σ_γ. This discrete approximation introduces persistent heterogeneity without adding a continuous state variable. Households with high γ_{ν_i} value consumption more and retire later; households with low γ_{ν_i} value leisure more and retire earlier.

Leisure is reduced both by health limitations and by the time required to work:

$$\ell_{ij} = 1 - \kappa(H_{ij}) - n_{ij}\, h(j), \qquad h(j) = h_0 + z(j - 55).$$

Here $h_0 \in (0, 1)$ is the baseline fraction of the time endowment lost when working at age 55, and z controls how that time cost changes with age. When $z < 0$, the time requirement of work decreases with age, capturing the transition toward part-time employment at older ages. The health time costs satisfy $\kappa(1) = 0$ and $\kappa(2), \kappa(3) \geq 0$. Fair and poor health therefore reduce available leisure even when not working. Health affects the labor supply decision through two distinct margins: the time endowment κ reduces leisure regardless of work status, while the work cost $\psi(H)$ raises the direct disutility of employment in poor health.

Bequests. Following De Nardi (2004) and Lockwood (2018), if death occurs at the end of age j, the household derives warm-glow utility from next-period assets:

$$v(a) = \left(\frac{\phi_b}{1 - \phi_b}\right)^{\sigma_b} \frac{\left(\frac{\phi_b}{1-\phi_b}\underline{c}_b + a\right)^{1-\sigma_b}}{1 - \sigma_b}, \qquad \phi_b \in [0, 1),$$

with $v(a) = 0$ when $\phi_b = 0$. The parameter ϕ_b governs the strength of the bequest motive. The shifter $\underline{c}_b > 0$ controls the asset level at which bequests become quantitatively important: higher values flatten the marginal utility of bequests at low wealth. Bequest curvature is tied to period-utility curvature through

$$\sigma_b = \eta_b \sigma,$$

where $\eta_b > 0$ is estimated jointly with the other household parameters. This specification allows the bequest margin to be more or less curved than the consumption-leisure margin. When $\eta_b < 1$, bequest utility is less concave than flow utility, so that the marginal utility of bequests declines more slowly with wealth. Saving for bequests is then a luxury good: permanently richer households choose to leave disproportionately larger estates, generating the concavity of consumption in permanent income documented by Straub (2019). Straub (2019) estimates a permanent-income elasticity of consumption of approximately 0.7, significantly below the unit elasticity implied by $\eta_b = 1$. This non-homotheticity is quantitatively important for matching the cross-sectional wealth distribution across permanent-income groups, as we show in Section 5.

3.3 Health, Survival, and Earnings

Following the procedure described in French and Jones (2011), health follows a finite-state Markov process estimated outside the structural model:

$$\Pr(H_{i,j+1} = h' \mid H_{ij} = h, \theta_i, j) = \Pi_H(h' \mid h, \theta_i, j).$$

The transition matrix varies with age and permanent earnings type. This allows poorer households to face systematically worse health dynamics, as in the data. We assume that health is exogenous. This is a reasonable simplification because we focus on older individuals who have already shaped their health and lifestyle. It is not clear ex ante whether this causes us to understate or overstate the labor-supply response, since endogenous health investment could either substitute for or complement continued work.

Conditional survival probabilities are

$$s_j(h, \theta_i) = \Pr(\text{survive to age } j+1 \mid H_{ij} = h, \theta_i, j).$$

Mortality risk therefore depends on both current health and permanent earnings type. Health and survival are estimated from HRS data and held fixed throughout structural estimation.

Labor income is deterministic conditional on age, permanent type, and health. If the household works at age j, annual earnings are

$$y_j(\theta_i, H_{ij}).$$

Because there is less earnings uncertainty during the late career than during the working years, and because the model focuses on the participation margin, we simplify the earnings process by making it deterministic conditional on the state. These profiles are estimated outside the dynamic problem. Because the model has only an extensive labor-supply choice, $y_j(\theta_i, H_{ij})$ should be interpreted as annual earnings conditional on working, not as an hourly wage schedule.

3.4 Social Security

The model tracks a reduced-form Social Security earnings state $\bar{A}_{ij}$, measured in monthly AIME units. This state evolves only before claim and only if the household works. Specifically,

$$\bar{A}_{i,j+1} = \begin{cases} \bar{A}_{ij} + \dfrac{1}{35} \max\left\{ \dfrac{\min\{y_j(\theta_i, H_{ij}), \bar{y}\}}{12} - \bar{A}_{ij}, 0 \right\} & \text{if } q_{ij} = 0 \text{ and } n_{ij} = 1, \\ \bar{A}_{ij} & \text{otherwise,} \end{cases} \tag{4}$$

where $\bar{y}$ is the taxable maximum. Working can replace low historical earnings years in the AIME calculation, but only subject to the taxable cap and only before claim. The law of motion is a reduced-form approximation that summarizes the incremental benefit consequences of late-life work through a one-dimensional sufficient statistic rather than reconstructing the full earnings history. Once claim occurs, $\bar{A}_{ij}$ is fixed.

Claiming. Claiming is a one-time, irreversible decision. An unclaimed household with $q_{ij} = 0$ may initiate benefits at any age $j \in \{62, \ldots, 70\}$ by setting $\iota_{ij} = 1$. At age 70, any remaining unclaimed household is forced to claim. Once benefits begin, the realized claim age is stored permanently:

$$q_{i,j+1} = \begin{cases} q_{ij} & \text{if } q_{ij} > 0, \\ j & \text{if } q_{ij} = 0 \text{ and } \iota_{ij} = 1, \\ 0 & \text{otherwise.} \end{cases}$$

The household's optimal claim timing depends on the tradeoff between receiving a smaller benefit sooner — with protection from future cuts on the already-received payments — and waiting for a larger benefit base that remains exposed to policy risk for longer.

Benefits and the Common Cut. Retired-worker benefits follow the standard U.S. formula. Monthly Primary Insurance Amounts are

$$\text{PIA}(\bar{A}) = \begin{cases} 0.90\,\bar{A} & \text{if } \bar{A} \le \kappa_1, \\ 0.90\,\kappa_1 + 0.32(\bar{A} - \kappa_1) & \text{if } \kappa_1 < \bar{A} \le \kappa_2, \\ 0.90\,\kappa_1 + 0.32(\kappa_2 - \kappa_1) + 0.15(\bar{A} - \kappa_2) & \text{if } \bar{A} > \kappa_2, \end{cases} \tag{5}$$

where $(\kappa_1, \kappa_2) = (761, 4586)$ in monthly dollars. Claim-age adjustments relative to the Full Retirement Age $j_F = 67$ are

$$\eta(q) = \begin{cases} 1 - \frac{5}{9} \times \frac{\min\{12(j_F - q), 36\}}{100} - \frac{5}{12} \times \frac{\max\{12(j_F - q) - 36, 0\}}{100} & \text{if } q < j_F, \\ 1 + \frac{2}{3} \times \frac{12(q - j_F)}{100} & \text{if } q \geq j_F. \end{cases} \tag{6}$$

The scheduled annual benefit is $\mathcal{B}_q(\bar{A}) = 12 \times \eta(q) \times \text{PIA}(\bar{A})$. If the common benefit cut has occurred ($\omega_j = 1$), all current Social Security payments are scaled down by the factor $(1 - \delta_{\text{cut}})$, with $\delta_{\text{cut}} \in (0, 1)$. Actual benefit income in period j is

$$\tilde{b}_{ij} = \begin{cases} (1 - \delta_{\text{cut}}\, \omega_j)\mathcal{B}_{q_{ij}}(\bar{A}_{ij}) & \text{if } q_{ij} > 0, \\ (1 - \delta_{\text{cut}}\, \omega_j)\mathcal{B}_j(\bar{A}_{ij}) & \text{if } q_{ij} = 0 \text{ and } \iota_{ij} = 1, \\ 0 & \text{otherwise.} \end{cases} \tag{7}$$

The first branch applies to households that have already claimed. The second branch applies to households that initiate a new claim in the current period: they receive benefits immediately at claim age j, consistent with the within-period timing above. Benefits already received in prior periods are not clawed back. This asymmetry is the source of the claiming margin identified in Section 2: early claiming shortens the household's effective duration of exposure to the common cut, at the cost of a permanently lower benefit base.

3.5 Solvency Risk

The policy-risk block is the key departure from standard retirement models. It connects the stylized mechanism in Section 2 to the quantitative life-cycle problem.

The common cut regime. The aggregate policy state $\omega_j \in \{0, 1\}$ is common to all households. In the promised regime ($\omega_j = 0$), benefits are paid as scheduled. In the cut regime ($\omega_j = 1$), all Social Security payments are permanently scaled by $(1 - \delta_{\text{cut}})$. The cut regime is absorbing: once $\omega_j = 1$, it remains so thereafter. This common stochastic cut is a stylized summary of the reform risk that households face. It is not a literal institutional forecast; rather, it is a reduced-form wedge that captures the possibility of trust-fund depletion, a broad solvency reform, or any legislative change that lowers benefits for all recipients once it takes effect. The cut size δ_{cut} is calibrated from the Social Security Trustees Report, which projects that

after trust-fund exhaustion payroll-tax revenue would cover approximately 75–83 percent of scheduled benefits, implying $\delta_{\text{cut}} \in [0.17, 0.25]$.

Common cut hazard. The transition probability from the promised to the cut regime is pinned by the observed solvency-risk index G_t described in Section 4.1. Because the index is a calendar-year object, the cut hazard for a cohort-c household at age j is evaluated at calendar year $t = c + j$:

$$p_j^c \equiv \Pr(\omega_{j+1} = 1 \mid \omega_j = 0) = h(G_{c+j}; \alpha_0, \alpha_1), \tag{8}$$

where h is a logistic function,

$$h(G; \alpha_0, \alpha_1) = \frac{1}{1 + \exp\big(-(\alpha_0 + \alpha_1 G)\big)},$$

and (α_0, α_1) are calibrated parameters. Higher fiscal stress raises the cut hazard. All households within the same cohort and calendar year share the same p_j^c. This common-belief specification avoids the computational cost of a household-specific belief state. Because different cohorts reach the same age in different calendar years, they face different cut-hazard paths: a cohort turning 62 in 2008 faces the pre-crisis solvency environment, while a cohort turning 62 in 2020 faces the post-pandemic environment. This cross-cohort variation in the realized G_t path at each age is the source of identifying content for (α_0, α_1).

The logistic form is a tractable mapping that satisfies the qualitative restriction $h' > 0$ from the theory: the cut hazard is increasing in fiscal stress. The parameter α_0 governs the baseline level of the hazard and α_1 governs its sensitivity to fiscal stress. Section 5 describes how they are disciplined.

Two margins of response. This block generates the two margins isolated in Section 2. First, a *precautionary work-and-saving* channel: when the cut hazard p_j rises, the marginal value of private assets increases because the household faces a larger expected shortfall in future benefit income. This strengthens the incentive to work and save. Second, a *claiming front-loading* channel: when p_j rises, the option value of waiting to claim falls because a larger benefit base is also more exposed to the common cut. Early claiming shortens the duration of exposure at the cost of a permanently lower benefit. The model is therefore built to allow retirement timing and claiming timing to move in different directions in response to the same solvency-risk shock.

3.6 Budget Constraint

The household budget constraint is

$$c_{ij} + a_{i,j+1} = R\, a_{ij} + y_j(\theta_i, H_{ij})\, n_{ij} + \tilde{b}_{ij}, \qquad a_{i,j+1} \geq 0, \tag{9}$$

where $R = 1 + r$ is the gross return on assets and $\tilde{b}_{ij}$ is the actual benefit income defined in (7). We set $r = 0.0288$, so the annual gross return is 1.0288. The borrowing limit is zero. For tractability, the budget constraint abstracts from an explicit payroll-tax wedge and aggregate wage shocks. Because the model focuses on the participation margin and Social Security benefit incentives, this abstraction is unlikely to bias the late-life employment response in a systematic direction.

3.7 Recursive Formulation

Let $V_j^c(\mathbf{x}_{ij})$ denote the value function at age j for a household in cohort c. The cohort superscript indexes the exogenous G_t path that the household faces; it is not a state variable but a parameter of the backward-induction problem. Because claiming is forced at age 70 and $J = 90$, all households have already claimed by the terminal age. At the terminal age,

$$V_J^c(\mathbf{x}_{iJ}) = \max_{n_{iJ} \in \{0,1\},\, a_{i,J+1} \geq 0} \left\{ u(c_{iJ}, \ell_{iJ}, n_{iJ}) + v(a_{i,J+1}) \right\},$$

subject to (9), with $q_{iJ} > 0$ guaranteed. For ages $j < J$, the household solves

$$\begin{aligned} V_j^c(\mathbf{x}_{ij}) = \max_{\substack{n_{ij} \in \{0,1\},\, \iota_{ij}, \\ a_{i,j+1} \geq 0}} \Big\{ & u(c_{ij}, \ell_{ij}, n_{ij}) \\ & + \beta\, s_j(H_{ij}, \theta_i)\, \mathbb{E}_j[V_{j+1}^c(\mathbf{x}_{i,j+1})] \\ & + \beta(1 - s_j(H_{ij}, \theta_i))\, v(a_{i,j+1}) \Big\}, \end{aligned} \tag{10}$$

where consumption is pinned down by (9) and the claiming choice ι_{ij} is active only for unclaimed households at ages 62–70 (it is zero otherwise). In the promised regime ($\omega_j = 0$), the expectation $\mathbb{E}_j$ integrates over next-period health and over the binary regime transition: with probability $p_j^c = h(G_{c+j})$ the household enters the cut regime, and with probability $1 - p_j^c$ it remains in the promised regime. In the cut regime ($\omega_j = 1$), the regime is absorbing and the expectation is over next-period health alone. Because the cut hazard depends on G_{c+j}, two cohorts at the same age j but in different calendar years face different transition probabilities. The model is

solved backward separately for each cohort.

The remaining state transitions are as follows. Claiming evolves as described above. The Social Security earnings state follows (4). Health evolves according to Π_H, and survival is governed by $s_j(H_{ij}, \theta_i)$. Permanent earnings type θ_i and preference type ν_i are fixed. The only endogenous state transitions come through savings, work, and claiming.

The household therefore faces four operative margins. Working raises current resources and, before claim, can also raise future benefits through (4). Saving shifts resources into future ages and into the bequest margin. Claiming converts future benefit promises into current protected income. Solvency risk tilts the household toward earlier claiming and stronger precautionary saving, as formalized in Section 2. The interaction between claiming and work is the central object of the quantitative analysis.

Estimation benchmark. The estimation benchmark described in Section 5 first solves and estimates a restricted version of this problem in which claiming follows a deterministic type-health schedule $\bar{q}(\theta_i, H_{ij})$ and the policy-risk block is shut down ($p_j = 0$ for all j). These restrictions keep the estimation problem computationally tractable while matching the employment, wealth, and bequest moments that discipline the household preference parameters. The counterfactual exercises in Section 7 then activate the full model — endogenous claiming and the solvency-risk block — using the estimated preference parameters. Section 5 describes the additional moments that identify the solvency-risk parameters $(\delta_{\text{cut}}, \alpha_0, \alpha_1)$ and the claiming response.

4 Data and Measurement

Official Social Security solvency risk is persistent, economically salient, and revised in discrete steps. Since 1995, every annual Trustees Report has projected OASDI trust-fund depletion within roughly 35 to 40 years, the date has been revised in 18 of the past 30 reports, and Congress has enacted three statutory payroll-tax adjustments. For households and policymakers, the threat of benefit cuts at depletion, the realization of payroll-tax changes, and annual revisions to the projected depletion date are therefore part of the environment they plan against.

However, no single statistic is a sufficient summary of the fiscal state that households actually face. The 75-year actuarial balance is, by construction, the present value of OASDI tax income minus the present value of cost (with a terminal reserve adjustment) over a fixed 75-year window, expressed as a percent of the present value of taxable payroll over the same

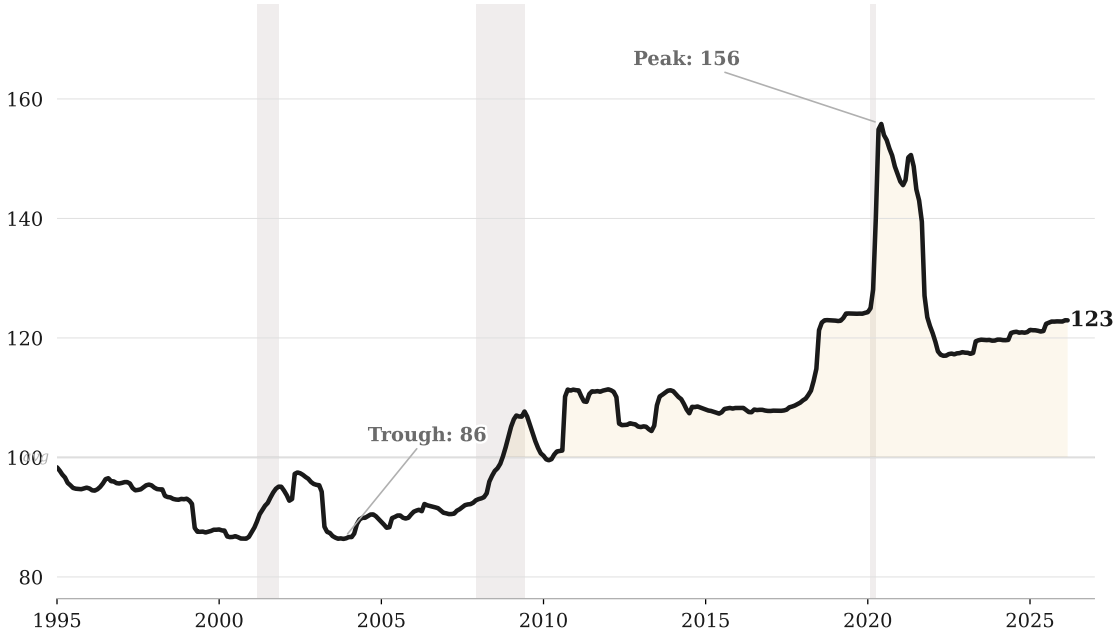

Figure 2: OASDI Solvency Risk Index

Notes: The figure plots the monthly OASDI solvency-risk index normalized so that its 1995–2019 average equals 100 and one standard deviation equals 10 index points. The dashed line marks the baseline average. The gold-shaded area highlights periods when the index exceeds the baseline. Shaded vertical bars are NBER recession dates. Labeled values mark the peak, trough, and latest observation. Source: authors' construction from SSA Trustees Reports, CBO Social Security baselines, and FRED.

window[1]. Because it integrates and discounts annual cash flows, tow identical actuarial balances can imply very different depletion years. The statistic implies the size of the long-run gap but not they date at which the trust fund reaches depletion.

Figure 2 plots the monthly OASDI solvency-risk index normalized to a 1995–2019 average of 100. The series is built from official SSA Trustees and CBO materials and rises when the financing gap worsens, or the depletion horizon shortens. It therefore summarizes and captures the official fiscal environment facing households, not media attention or generic macro uncertainty.

First, Figure 2 shows that Social Security solvency risk is elevated relative to its historical norm, but not monotone. The OASDI index is normalized to 100 over 1995–2019. It averages 91.5 in 2006–2007, rises to 101.2 in 2008–2009 during the Global Financial Crisis, jumps much higher during the pandemic, averaging 142.9 in 2020–2021 and peaking at 155.8 in June 2020,

[1]The exact formula is

$$\text{Actuarial Balance}_{75} \equiv \frac{\text{PV(income)} - \text{PV(cost)} - \text{End-of-period target reserve} + \text{Initial trust fund}}{\text{PV(taxable payroll)}}$$

and remains elevated at 122.1 on average in 2025–2026, with a latest value of 122.9 in March 2026. The main fact is therefore not simply that the current level is above average, but that large macroeconomic disruptions and subsequent official re-estimates can move the solvency environment into a distinctly worse regime.

Second, the index evolves through discrete revaluations rather than a smooth upward drift. It remains near one level for long stretches, then moves abruptly upward around episodes such as the 2008–2009 financial crisis, the pandemic, and later Trustees and CBO updates, before settling at a new plateau. In our construction, those jumps come from two sources. Official SSA and CBO releases reprice the long-run financing gap and depletion horizon, while sharp changes in the payroll-tax base and labor market raise near-term revision risk between releases. For our purposes, the key implication is that households face solvency risk as a sequence of identifiable news shocks rather than as a slowly varying background condition.

4.1 Fiscal-Risk Measurement

We construct two monthly series: an OASI legal-headline index and an OASDI public-salience index. Figure 2 reports the OASDI series. Methodologically, the construction combines official fiscal accounting, a small state-space factor model in the spirit of Stock and Watson (1988) and Doz et al. (2012), and a text layer that draws on audited dictionary methods à la Baker et al. (2016), contextual risk language in the spirit of Hassan et al. (2019), and sentence-embedding classification following Reimers and Gurevych (2019). The distinction between solvency stress and solvency uncertainty follows the logic of Jurado et al. (2015).

The construction proceeds in four stages: (i) assembling a real-time source panel, (ii) separating the signal into a stress component and an uncertainty component, (iii) extracting one latent factor from each component, and (iv) combining them into a single rescaled index.

Sources and real-time timing. The inputs are every annual SSA Trustees Report from 1995 through the latest release, every CBO Social Security baseline over the same window, and three monthly FRED macro series: total non-farm payrolls, average private hourly earnings, and the unemployment rate. For each SSA and CBO document we record its release date and assign it to a monthly timeline using a simple rule: in any given month, the numbers are those from the most recent report published on or before that month. The values stay flat until the next report comes out and updates them. This way our series reflects exactly what a household or researcher could have known at the time, never using information that had not yet been published.

Stress component. The stress block combines five official solvency levels with one higher-frequency macro summary, for each basis (*OASI* and *OASDI*). From each Trustees vintage we parse (a) the 75-year actuarial balance, (b) the open-group unfunded obligation, (c) the years until trust-fund depletion, (d) the payable-benefits shortfall at depletion — the share of scheduled benefits that current-law revenue cannot cover after exhaustion — and (e) a trust-fund weakness measure built from the minimum projected reserve ratios over the next 10 and 25 years. To these we add a monthly macro-stress indicator that summarizes payroll-base deterioration: it combines the year-over-year decline in total wage-bill growth, constructed as the product of employment and hourly-earnings growth, with the level of the unemployment rate. All six series are signed so that a higher value always means a more adverse environment and then standardized within series before factor extraction.

Uncertainty component. The uncertainty block captures disagreement about the same objects rather than their level. It combines (a) the low–high scenario spread in the Trustees depletion year, (b) the stochastic depletion spread from the Trustees probability simulations, (c) a 12-month rolling standard deviation of cross-vintage revision shocks, where each revision shock is the mean of the standardized absolute release-to-release changes in the actuarial balance, the unfunded obligation, years-to-depletion, and the scenario width, (d) the absolute gap between the SSA and CBO depletion-year projections, and (e) a macro revision-risk measure built from short-run changes and rolling volatility of the macro-stress series. Following Jurado et al. (2015), the purpose of this block is to isolate uncertainty about future solvency outcomes rather than just the current level of the fiscal gap. A sixth input, an optional sentence-level text uncertainty score computed from Trustees, CBO, and congressional hearing texts, enters the block only if a preregistered audit gate passes.

Text analysis. The text pipeline is intentionally hybrid. We first build sentence-level candidates from SSA, CBO, and congressional documents using program and solvency terms, and compute dictionary-style and contextual proximity features in the spirit of Baker et al. (2016) and Hassan et al. (2019). We then validate those sentences against a labeled audit sample using a local sentence-embedding classifier built from sentence embeddings, auxiliary text features, and calibrated linear classifiers, following the sentence-representation logic of Reimers and Gurevych (2019). In the current implementation, the released monthly text measure should be interpreted as an audited hybrid uncertainty score rather than as a pure transformer probability. It is admitted into the uncertainty block only if the audit gate passes on reviewed-label coverage, inter-annotator agreement, holdout precision and recall, and calibration.

Two-block factor model. For each basis, we fit two independent single-factor state-space models, one to the stress block and one to the uncertainty block, following the logic of Stock and Watson (1988) and Doz et al. (2012). Each raw input series x_{it} is first standardized within series:

$$z_{it} = \frac{x_{it} - \bar{x}_i}{\sigma_i},$$

where $\bar{x}_i$ and σ_i are the full-sample mean and standard deviation of series i. Within each block, the standardized observables are modeled as noisy measurements of a single latent factor f_t:

$$\text{State:} \quad f_t = \phi\, f_{t-1} + \varepsilon_t, \qquad \varepsilon_t \sim N(0, q), \tag{11}$$

$$\text{Observation:} \quad z_{it} = \lambda_i\, f_t + u_{it}, \qquad u_{it} \sim N(0, r_i), \tag{12}$$

where ϕ is the AR(1) persistence of the factor, λ_i is the loading of series i on the common factor, and r_i is the idiosyncratic noise variance. Estimation proceeds by iterating three times between (a) estimating the loadings λ_i and noise variances r_i by regressing each z_{it} on the current factor estimate, (b) estimating ϕ and q from an AR(1) regression of the factor on its lag, and (c) re-extracting the factor via the Kalman smoother with the updated parameters. Each factor is sign-identified so that a more adverse observable (a worse actuarial balance for stress, a wider scenario spread for uncertainty) raises the factor. We estimate the two blocks separately rather than pooling them because the level of solvency risk and uncertainty about that level are economically distinct objects that move at different frequencies and can differ. The 2020 period, for example, shows larger movements in uncertainty than in level stress.

Figure 3 decomposes the headline index into its two latent factors. The stress component is a slow-moving staircase that ratchets upward with each Trustees revision and rarely reverts. The uncertainty component is spikier: it surges during the Global Financial Crisis and especially the COVID-19 pandemic, then partially mean-reverts. The 2020 spike in the headline is almost entirely an uncertainty event, while the post-2022 elevation reflects a convergence of both components at historically high levels.

Headline index. Each extracted factor is re-standardized against the 1995–2019 baseline period to produce f_t^{stress} and f_t^{unc}, each with baseline mean zero and unit variance. The headline factor is a fixed convex combination:

$$G_t^{\text{headline}} = 0.70 \times f_t^{\text{stress}} + 0.30 \times f_t^{\text{unc}}.$$

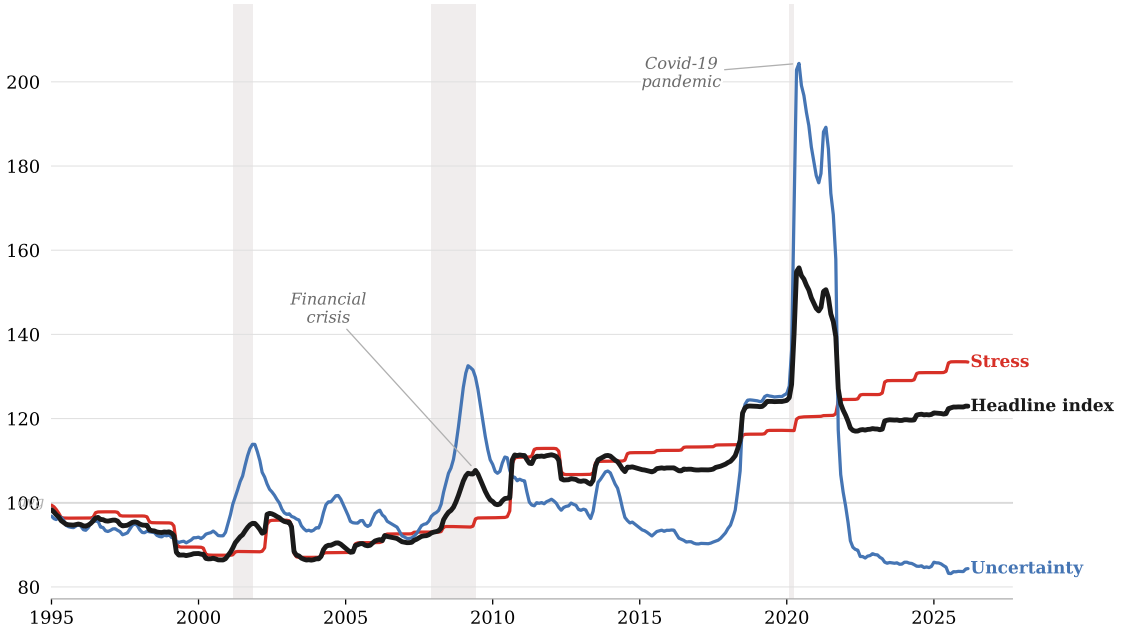

Figure 3: OASDI Solvency Risk Index: Stress and Uncertainty Components

Notes: The figure decomposes the headline OASDI solvency-risk index (black) into its stress (red) and uncertainty (blue) components. Each series is rescaled so that its 1995–2019 average equals 100 and one standard deviation equals 10 index points. The stress component captures the level of fiscal deterioration (actuarial balance, depletion horizon, payable shortfall); the uncertainty component captures disagreement and revision volatility (scenario spreads, SSA–CBO gaps, cross-vintage shocks). The headline index is $0.70 \times \text{Stress} + 0.30 \times \text{Uncertainty}$. Shaded vertical bars are NBER recession dates. Source: authors' construction from SSA Trustees Reports, CBO Social Security baselines, and FRED.

The weighting encodes a prior that households react primarily to observable deterioration in the *level* of solvency, with cross-forecaster disagreement and revision volatility acting as a secondary amplifier. For the structural model, we use G_t in its standardized form (approximately mean zero and unit standard deviation over 1995–2019). For presentation, the index is linearly rescaled so that its 1995–2019 mean equals 100 and its standard deviation equals 10, producing the series in Figure 2. Appendix Table 7 reports the annual values of G_t used in the structural estimation.

5 Calibration

5.1 Estimation Strategy

The estimation proceeds in three steps, following the general approach of Gourinchas and Parker (2002) and French (2005). In the first step, we estimate or calibrate parameters that can be cleanly identified without solving the structural model: the entry distribution, initial asset and

AIME supports, earnings profiles, health transitions, survival probabilities, and the cut size δ. In the second step, we estimate the preference vector Θ_H by the method of simulated moments, using 25 cross-sectional moments pooled across all HRS cohorts observed in 1995–2008. In the third step, we fix $\hat{\Theta}_H$, activate endogenous claiming and the solvency-risk block, and estimate the solvency parameters (α_0, α_1) on the 1943 birth cohort.

The rationale for the split is identification. The preference parameters are identified from cross-sectional variation in employment, wealth, and decumulation by age, health, and permanent-income group. Pooling across cohorts gives a large sample (∼14,000 person-wave observations) and precise moments. The solvency parameters, by contrast, are identified from how claiming and employment respond to time-varying fiscal stress. That identification requires a clean institutional environment in which the only time-varying object is the solvency-risk index G_t. The 1943 birth cohort provides exactly this: its members face the permanent post-transition Social Security rules (FRA = 66, DRC = 8% per year, no Retirement Earnings Test above FRA) throughout their claiming window (ages 62–70, calendar years 2005–2013), and the solvency index exhibits substantial variation over that window — from the pre-crisis baseline through the financial crisis and into the post-crisis period.

5.2 External Calibration

Table 1 lists the objects estimated or set outside the structural model and held fixed during both estimation steps. Panel A of Table 4 lists fixed institutional quantities. Panel B lists the externally calibrated inputs.

Permanent types and initial conditions. The permanent type θ is a fixed earnings-type label constructed from the HRS before the model is solved. For each respondent, we collect all valid earnings observations at ages 54–56, deflate them to constant 2010 dollars using the CPI-U, and compute the person-level mean. Weighted quintile cutoffs define $N_\theta = 5$ permanent types. Each respondent receives one fixed θ_i, and the same assignment is used in every subsequent calibration step. We then estimate the joint entry distribution $p(\theta, H_{55})$ from the pooled HRS age-54–56 sample.

Conditional on θ, we calibrate the age-55 entry conditions outside the structural model. Initial assets are drawn from quintile-specific distributions centered on the HRS medians reported in Appendix Table 9. The model's initial Social Security state is drawn from a quintile-specific AIME-equivalent proxy. Because the public HRS does not directly report accrued age-55 AIME, we back it out from prospective own-benefit wealth at FRA in the

Table 1: External Calibration Inputs

Input	Construction	Identifies
Entry type-health distribution	HRS respondents ages 54–56, with fixed earnings quintiles from mean real earnings at ages 54–56	$p(\theta, H_{55})$
Initial asset and AIME supports	Quintile-specific HRS medians and within-quintile dispersion estimates	$a_{55} \mid \theta,\ \bar{A}_{55} \mid \theta$
Earnings profiles	HRS age-health earnings profiles by permanent-income quintile	$y_j(\theta, H)$
Health transitions and survival	Annualized HRS transition matrix and survival probabilities by age, health, and quintile	$\Pi_H,\ s_j$
Empirical observer profile	HRS interview-parity, age, health, and quintile observer weights over 1995–2008 interviews	observed-sample model moments
Consumption-weight heterogeneity	Fixed discrete approximation with $\sigma_\gamma = 0.067$ and $N_\gamma = 3$	γ_i type support
Cut size δ	Social Security Trustees Report: 77% payable at depletion	$\delta = 0.23$

Notes: All inputs are fixed before the household problem is solved. Model moments are constructed using the empirical HRS observer profile.

HRS Prospective Social Security Wealth file and map it into the model using the same PIA formula described in Section 3. We interpret $\bar{A}_{55}$ as the model's entry SS state, not as a direct administrative measure. Households enter the model before claiming, so $q_{55} = 0$. These entry objects are fixed before the preference parameters are estimated.

Table 2 reports the joint distribution of income quintile and health at entry. Health and income are strongly correlated: households in the top quintile enter later life with both higher earnings capacity and better health than households in the bottom quintile.

Health transitions and survival. Following the procedure described in French and Jones (2011), we estimate the health transition matrix $\Pi_H(h' \mid h, \theta, j)$ and the survival probabilities $s_j(h, \theta)$ from HRS panel data using all available waves (waves 3–15, 1996–2020). We use the full sample rather than restricting to the 1998–2008 preference-estimation window because health transitions are biological processes that do not depend on Social Security rules, and the wider age support (55–79 vs. 55–65) is needed for stable estimation of the smooth age profiles. Self-reported health is collapsed from five categories to the model's three states: $H = 1$ (excellent or very good), $H = 2$ (good or fair), and $H = 3$ (poor). We estimate separate gender-specific models for the two-year death probability and for next-interview health transitions, using smooth weighted logit and multinomial logit specifications with age and permanent-type controls. Two-year fitted probabilities are annualized assuming a constant hazard within each HRS interview

Table 2: Joint Distribution of Income and Health at Age 55

	Health Status			
Income Quintile	Good ($H = 1$)	Fair ($H = 2$)	Poor ($H = 3$)	$p(\theta_k)$
Q1 (below \$19,419)	0.0794	0.0955	0.0110	0.1859
Q2 (\$19,419–\$34,336)	0.0942	0.1028	0.0068	0.2038
Q3 (\$34,336–\$52,115)	0.1032	0.0916	0.0055	0.2003
Q4 (\$52,115–\$79,401)	0.1239	0.0789	0.0014	0.2042
Q5 (above \$79,401)	0.1406	0.0639	0.0013	0.2058
$p(H_{55})$	0.5412	0.4327	0.0260	1.0000

Notes: Weighted joint distribution of permanent-income quintile and model health status. Permanent-income types are assigned from mean real earnings at ages 54–56 using the full HRS sample (waves 3–15, 1996–2020) for precision of the quintile cutoffs. The joint distribution itself is computed from HRS respondents aged 54–56 across the same waves. Health mapping: Good = HRS excellent/very good; Fair = HRS good/fair; Poor = HRS poor. Sample: 14,878 person-wave observations (11,430 unique respondents). Health transitions and survival are estimated on the full HRS sample (waves 3–15, 1996–2020) for age-support coverage; see text for justification.

interval. The gender-specific annual objects are then pooled using the observed HRS gender shares within each (θ, h) cell. These health and survival objects are held fixed in the structural estimation. As an external check, Appendix Table 10 compares the implied mortality profiles with SSA period life tables.

Age-earnings profiles. Following French (2005), we estimate the deterministic labor-income component $y_j(\theta, H)$ from annual earnings among HRS respondents in a worker sample designed to match the model's single work state. The worker sample requires positive earnings, positive survey weight, non-missing health, and at least 500 annual hours. For each gender and permanent-income type, we fit a penalized regression for log annual earnings on a cubic age spline with period and cohort controls. We then estimate age-varying health deviations around the type-specific base curve, subject to the monotone ordering $y_j(\theta, \text{Good}) \geq y_j(\theta, \text{Fair}) \geq y_j(\theta, \text{Poor})$. The gender-specific profiles are pooled to the model's (θ, H) state space using age-specific HRS gender shares. Appendix A.4 gives the exact sample definition, smoothing equations, and pooling rule. Appendix Figure 8 validates the pooled age shape against CPS ASEC data.

5.3 Simulation

With the external objects described above held fixed, we estimate the preference vector Θ_H by the method of simulated moments (SMM; McFadden, 1989; Pakes and Pollard, 1989). The model used in this step imposes a deterministic claiming schedule and shuts down the

solvency-risk block ($p_j^c = 0$). The moments are pooled across all HRS cohorts observed in 1995–2008.

Let

$$\Theta_H = (\beta, \sigma, \gamma, \psi_0, \delta, h_0, z, \kappa_2, \kappa_3, \phi_b, \underline{c}_b, \eta_b)$$

denote the twelve-dimensional household parameter vector, where $\eta_b \equiv \sigma_b/\sigma$ is the bequest-curvature ratio. The SMM estimator minimizes the quadratic distance between data moments and simulated model moments:

$$\hat{\Theta}_H = \arg\min_{\Theta_H} (m^{\text{data}} - m^{\text{model}}(\Theta_H))' W (m^{\text{data}} - m^{\text{model}}(\Theta_H)), \tag{13}$$

where m^{data} is the 25-dimensional vector of targeted HRS moments described in Table 3, $m^{\text{model}}(\Theta_H)$ is its simulated counterpart, and W is the weighting matrix. We weight by the regularized inverse of the full bootstrap covariance matrix of the data moments. This gives more weight to moments that are measured more precisely.

Procedure. The estimation can be summarized as follows. First, we fix all external objects: health transitions, survival, earnings profiles, the entry distribution, the claiming schedule, and the empirical observer profile. Second, for each candidate parameter vector Θ_H, we solve the household dynamic program by backward induction from age $J = 90$ to the entry age 55. Third, we simulate $N_{\text{sim}} = 50{,}000$ households forward from the age-55 joint type-health distribution. Fourth, we construct model moments on the same observed-sample definition as the HRS target file, using the empirical observer profile to weight the simulated panel. Fifth, we compute the SMM objective (13) and update the parameter vector. Sixth, we iterate until the objective is minimized. The search uses a bounded coordinate-descent pass followed by a bounded Nelder–Mead simplex refinement (Nelder and Mead, 1965).

Targeted moments. The problem is over-identified: 25 moments and 11 parameters. The target vector has three blocks. The labor block contains eight employment rates over ages 55–61, 62–64, 65–69, 70–74, 75–79, 80–84, 85–89, and 90–94, together with the nonwork hazard at age 62 and the good-fair and good-poor employment gaps over ages 62–70. The savings-bequest block contains median wealth at ages 65–69, median decumulation from ages 65–69 to 75–79, median wealth at death, and the interquartile range of wealth at death. The remaining ten moments are permanent type-specific wealth medians at ages 65–69 and at death for each permanent-income quintile. Table 3 lists the full target vector; Table 4 reports the parameter

vector with admissible bounds.

Table 3: Targeted Moments: Preference Estimation (Pooled HRS 1995–2008)

Moment	**Construction**	**Primary role**
Median wealth, ages 65–69	Median gross household wealth (2010$)	β
Median wealth decumulation, ages 65–69 to 75–79	Median wealth at 65–69 minus median wealth at 75–79 (2010$)	σ, η_b
Employment rate by age bin	Fraction working within each bin (8 moments: 55–61, 62–64, 65–69, 70–74, 75–79, 80–84, 85–89, 90–94)	γ, ψ_0, h_0, z
Nonwork hazard at age 62	Share not working at first post-62 interview among those working pre-62 (biennial bridge, pre 60–61, post 62–63)	ψ_0, h_0, z
Good–fair employment gap, ages 62–70	Employment difference: H=1 minus H=2	κ_2, δ
Good–poor employment gap, ages 62–70	Employment difference: H=1 minus H=3	κ_3, δ
Median wealth at death	Median last-observed gross household wealth before death (decedents)	$\phi_b, \underline{c}_b, \eta_b$
IQR of wealth at death	P75–P25 spread of last-observed gross household wealth before death (decedents)	$\underline{c}_b, \eta_b$
Median wealth, ages 65–69 by θ quintile	Median gross household wealth (2010$) within each permanent-income quintile	heterogeneity in savings by θ
Median wealth at death by θ quintile	Median last-observed wealth before death within each permanent-income quintile (decedents)	heterogeneity in bequests by θ

Notes: All moments from RAND HRS 1995–2008, survey-weighted. Wealth in 2010 dollars. θ quintiles are fixed by mean real earnings at ages 54–56. Model moments are constructed on the same observed-sample definition as the data by using the empirical HRS observer profile in simulation and moment aggregation. The weighting matrix is the regularized inverse of the full bootstrap covariance of the 25 targeted moments.

Identification. Each moment block disciplines a distinct subset of the preference vector. The employment-rate profile over ages 55–94 identifies γ, ψ_0, h_0, and z. The health-gap moments identify κ_2, κ_3, and δ. Median wealth at ages 65–69 and the decumulation moment discipline β and σ. The death-wealth moments identify the bequest parameters ϕ_b, $\underline{c}_b$, and η_b. The θ-specific wealth moments force the model to match the late-life wealth distribution across permanent-income groups, not just aggregate wealth levels.

The bequest-curvature ratio η_b deserves particular attention. Straub (2019) shows that in a general class of precautionary-savings models, the permanent-income elasticity of consumption is governed by σ_b/σ, which is exactly our η_b. When $\eta_b < 1$, bequests are a luxury good and richer households save disproportionately more. Straub estimates this elasticity at approximately 0.7 in PSID panel data, providing an independent benchmark. Importantly, Straub also finds that non-homotheticity in bequests alone accounts for only part of the empirical concavity; our estimated η_b may therefore absorb some of the omitted life-cycle channel.

Preference-stage validation. In this step, claiming behavior serves as an untargeted validation block. We compare the restricted model's simulated claiming profile with the 2010 HRS

Table 4: Model Parameters

Parameter	Description	Value / bounds	Source / role
Panel A Fixed institutional inputs			
(j_E, j_F, j_D)	Claiming ages	$(62, 66, 70)$	SSA rules for 1943 birth cohort
DRC	Delayed Retirement Credit	8% per year	SSA rules for 1943 birth cohort
κ_1, κ_2	PIA bend points (monthly)	\$761; \$4,586	SSA benefit formula
$\bar{y}$	Taxable maximum (annual)	\$106,800	SSA rules
δ_{cut}	Common benefit-cut size	0.23	Trustees Report: 77% payable at depletion
R	Gross asset return factor	1.0288	Fixed
$\underline{a}$	Borrowing limit	0	No unsecured borrowing
$\kappa(1)$	Good-health time cost	0	Normalization
Panel B Externally calibrated inputs (fixed during estimation)			
$p(\theta, H_{55})$	Entry type-health distribution	Table 2	HRS respondents ages 54–56
$a_{55} \mid \theta,\ \bar{A}_{55} \mid \theta$	Initial asset and AIME supports by quintile	Table 9	HRS-based initial-support files
$y_j(\theta, H)$	Annual earnings profile conditional on work	Appendix	External HRS earnings-profile estimation
$\Pi_H(h' \mid h, \theta, j),\ s_j(h, \theta)$	Health transitions and survival	Appendix	External HRS transition estimation
σ_γ	Std. dev. of persistent γ_i heterogeneity	0.067	Fixed
N_γ	Number of persistent γ_i support points	3	Fixed
Observer profile	Interview-parity / age / health / θ weights	Appendix	Empirical HRS observer file
Panel C Preference parameters (estimated in step 2, pooled HRS)			
β	Discount factor	[0.900, 1.000]	Wealth and bequest levels
σ	Curvature over the consumption-leisure composite	[1.5, 6.0]	Wealth level and decumulation
γ	Mean consumption weight in flow utility	[0.2, 0.8]	Employment profile and work-leisure tradeoff
ψ_0	Baseline work cost	[0.003, 0.300]	Employment profile and age-62 hazard
δ	Health work-cost gradient	[0.000, 0.200]	Health-employment gaps
h_0	Baseline time requirement of work	[0.010, 0.500]	Employment level
z	Age slope in time requirement	[−0.005, 0.100]	Late-life employment slope
κ_2	Fair-health time cost	[0.005, 0.150]	Good-fair employment gap
κ_3	Poor-health time cost	[0.010, 0.300]	Good-poor employment gap
ϕ_b	Bequest-share shifter	[0.000, 0.990]	Death-wealth moments
$\underline{c}_b$	Bequest location shifter	[\$10, \$200,000]	Death-wealth level and dispersion
$\eta_b = \sigma_b/\sigma$	Bequest-curvature ratio	[0.30, 0.99]	Curvature of the bequest margin
Panel D Solvency-risk parameters (estimated in step 3, 1943 cohort)			
α_0	Cut-hazard intercept	Estimated	Claiming-age distribution level
α_1	Cut-hazard sensitivity to fiscal stress	Estimated	Employment response to G_t variation

Notes: Panel A lists fixed institutional constants for the 1943 birth cohort. Panel B lists data-driven inputs loaded before estimation begins. Panel C reports the admissible bounds for the 12 preference parameters estimated in the second step from pooled HRS 1995–2008. Panel D lists the solvency-risk parameters estimated in the third step from the 1943 cohort. The cut size δ_{cut} is in Panel A because it is calibrated externally from the Trustees Report, not estimated.

cross-sectional claiming distribution at each exact age from 62 through 70 (Appendix Table 14). The claiming-age flow distribution is reserved for the solvency estimation step below.

5.4 Solvency-Risk Estimation

With $\hat{\Theta}_H$ fixed from the preference step, we activate endogenous claiming and the solvency-risk block and estimate the cut-hazard parameters (α_0, α_1) on the 1943 birth cohort.

Institutional environment. The 1943 birth cohort faces a fixed institutional regime throughout its claiming window. The Full Retirement Age is $j_F = 66$, fully phased in under the 1983 Amendments. The Delayed Retirement Credit is 8 percent per year of delay past FRA, which is actuarially fair. The Retirement Earnings Test applies only below FRA; the above-FRA test was repealed in 2000, well before this cohort reaches 62 in 2005. These rules are deterministic and known to the household. They enter the model through the actuarial adjustment $\eta(q)$ and the benefit computation $\tilde{b}_{ij}$ described in Section 3. Because no institutional rule changes during the cohort's claiming window, the only time-varying object in the household's problem is the cut hazard $p_j^c = h(G_{c+j}; \alpha_0, \alpha_1)$, which is driven by the observed solvency-risk index.

Identifying moments. The target vector contains two blocks of moments constructed from the 1943 birth cohort, and a narrow 1941–1945 band for sample size. The first block is the claiming-age flow distribution: the share of new claims at each exact age 62 through 70. The data counterpart is the SSA Annual Statistical Supplement (Table 6.B5.1), which reports the percentage distribution of retired-worker entitlements by age and year. We average the pooled men-and-women distribution over entitlement years 1998–2008, excluding disability conversions and renormalizing. In the model, the comparable object is the fraction of unclaimed households that initiate benefits at each age.

The second block is the employment response to solvency risk: employment rates at ages 62–64 and 65–69 for the 1941–1945 HRS birth-year, interacted with the level of the solvency-risk index at the interview date. These moments ask whether households observed during high-stress periods (e.g., 2008–2010) display different employment patterns than those observed during low-stress periods (e.g., 2005–2007), conditional on the same preferences and health. The cross-period variation in G_t within this narrow cohort band is the source of identifying content for α_1, the sensitivity of the cut hazard to fiscal stress.

Why this cohort. Three features make the 1943 birth cohort the natural estimation sample for the solvency block. First, its members face the permanent post-transition Social Security

rules, eliminating the confound between solvency-risk responses and institutional rule changes (the FRA increase and RET repeal) that affected earlier cohorts. Second, its claiming window (2005–2013) spans the most informative period of solvency-index variation: the pre-crisis baseline, the 2008–2009 financial crisis, and the post-crisis plateau. Third, the cohort is large enough in the HRS to construct the employment-by-solvency moments with reasonable precision when supplemented by adjacent birth years (1941–1945).

6 Model Fit and Main Quantitative Results

Table 5 reports the estimated preference vector $\hat{\Theta}_H$. The model is estimated from 25 moments constructed from the pooled HRS 1995–2008 panel, targeting the age profile of employment, the nonwork hazard at age 62, health-employment gaps, median wealth and decumulation, and wealth heterogeneity across permanent-income quintiles. All externally calibrated inputs — health transitions, survival, earnings profiles, the entry distribution, and the empirical observer profile — are held fixed at the values described in Section 5. We discuss parameter estimates, model fit to targeted moments, and validation against untargeted moments in turn.

6.1 Preference Estimates

Table 5: Preference Parameter Estimates

Parameter	Description	Estimate	Literature benchmark
β	Discount factor	0.996	0.97 (French and Jones, 2011)
σ	Curvature of the composite	2.40	3.81 (De Nardi et al., 2010)
γ	Consumption weight	0.302	0.36 (French and Jones, 2011)
ψ_0	Baseline work cost	0.004	—
δ	Health work-cost gradient	0.003	—
h_0	Baseline time requirement	0.020	—
z	Age slope in time requirement	−0.0004	—
κ_2	Fair-health time cost	0.088	—
κ_3	Poor-health time cost	0.213	—
ϕ_b	Bequest-share shifter	0.206	0.40 (Lockwood, 2018)
$\underline{c}_b$	Bequest location shifter	9,163	2,853 (Lockwood, 2018)
$\eta_b = \sigma_b/\sigma$	Bequest-curvature ratio	0.735	0.70 (Straub, 2019)

Notes: Estimated by the method of simulated moments from pooled HRS 1995–2008. The weighting matrix is the regularized inverse of the full bootstrap covariance of the 25 targeted moments. The model imposes a deterministic claiming schedule and shuts down the solvency-risk block ($p_j^c = 0$). Admissible bounds are reported in Appendix Table 11.

Discounting and curvature. We find that $\beta = 0.996$ and $\sigma = 2.40$. The discount factor is close to the values found by French and Jones (2011), who estimate $\beta = 0.97$, and Lockwood (2018), who estimates $\beta = 0.96$. The curvature parameter is lower than the $\sigma = 3.81$ reported in De Nardi et al. (2010), reflecting the different role σ plays in a Cobb–Douglas composite where it governs both consumption smoothing and the complementarity between consumption and leisure.

Labor supply. The consumption weight $\gamma = 0.302$ implies that consumption accounts for roughly 30 percent of the composite good, with the remaining 70 percent attributed to leisure. The baseline work cost $\psi_0 = 0.004$ is small, indicating that the disutility of work in good health operates primarily through the time channel rather than through a lump-sum utility penalty. The age slope $z = -0.0004$ is slightly negative, meaning the time requirement of work decreases modestly with age: $h(j) = h_0 + z \times (j - 55)$. At age 55, working consumes approximately 2 percent of the household's time endowment; by age 75, the time requirement falls to 1.2 percent. This low and gently declining time cost, combined with the high leisure weight, produces the gradual employment decline observed in the data.

Health costs. The fair-health time cost $\kappa_2 = 0.088$ and poor-health cost $\kappa_3 = 0.213$ imply that poor health reduces available time by roughly 2.4 times as much as fair health. These time costs operate symmetrically on workers and nonworkers. The health work-cost gradient $\delta = 0.003$ introduces a second, asymmetric channel: working in poor health carries an additional utility penalty of δ, with fair health receiving $\delta/2$. Together, the two channels produce a good–poor employment gap of 33.2 percentage points in the model, compared with 37.6 percentage points in the data, and a good–fair gap of 17.5 versus 12.8 percentage points.

Bequests. The bequest-share shifter is $\phi_b = 0.206$, the location shifter is $\underline{c}_b = \$9{,}163$, and the bequest-curvature ratio is $\eta_b = 0.735$. When $\eta_b < 1$, bequests are a luxury good and wealthier households save disproportionately more at the end of life. Straub (2019) estimates the permanent-income elasticity of consumption at approximately 0.7 in PSID panel data. Our estimate of $\eta_b = 0.735$ is close to the Straub benchmark, confirming that bequest preferences are substantially non-homothetic: permanently richer households choose to leave disproportionately larger estates, generating the concavity of consumption in permanent income. Lockwood (2018) estimates a bequest specification of the same functional form with $\phi_b = 0.40$ and $\underline{c}_b = \$2{,}853$; our bequest weight is lower and the location shifter is larger, reflecting the higher gross-wealth scale of our initial conditions.

6.2 Model Fit

Employment profile. Figure 4 compares the simulated age-employment profile with the HRS target. The model closely tracks the level and shape of the age-employment decline: 67.2 percent of simulated households work at ages 55–61 (data: 67.6%), falling to 51.7 percent at ages 62–64 (data: 46.1%) and 31.5 percent at ages 65–69 (data: 32.5%).

Although the model understates employment at ages 75 and above — where the simulated rate falls to 7.1 percent versus 12.0 percent in the data — it captures the steep transition into nonwork at age 62 and the continued decline through the 70s. The nonwork hazard at age 62 is 21.5 percent in the model and 25.1 percent in the data. The good–fair employment gap is 17.5 percentage points (model) versus 12.8 percentage points (data), and the good–poor gap is 33.2 versus 37.6 percentage points.

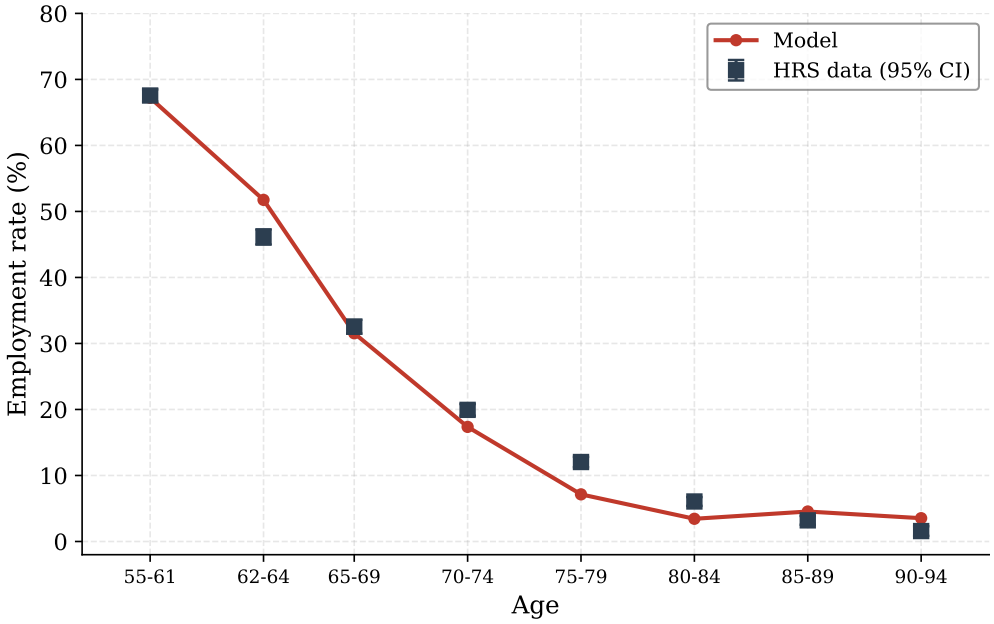

Figure 4: Employment Rate by Age: Model versus Data

Notes: Solid line: simulated employment rate from the estimated model (50,000 households, weighted by the empirical HRS observer profile). Markers: HRS target moments from pooled 1995–2008. Vertical bars: bootstrap 95% confidence intervals on the HRS moments.

Wealth accumulation and decumulation. Median household wealth at ages 65–69 is $265,621 in the model and $246,160 in the data. The median decumulation from ages 65–69 to 75–79 is $58,785 (model) versus $28,651 (data); the model overshoots decumulation, reflecting the combination of reduced late-life employment and a non-homothetic bequest motive that concen-

trates saving among the wealthy. Median wealth at death is $116,601 (model) versus $116,519 (data), closely matched. The interquartile range of wealth at death is $261,439 versus $305,029, indicating that the model captures the bulk of end-of-life wealth dispersion.

Quintile heterogeneity. Table 6 reports median wealth at ages 65–69 and at death for each permanent-income quintile. The model reproduces the strong positive gradient of wealth in permanent income: median wealth at ages 65–69 rises from $57,374 in the bottom quintile to $728,863 in the top quintile, compared with $130,604 to $724,077 in the data. The model understates wealth in the lower quintiles, where gross household wealth in the HRS includes housing equity that the model's liquid-asset framework does not fully capture. The top-quintile fit is close, with the model producing 101 percent of the data value.

Table 6: Wealth by Permanent-Income Quintile: Model versus Data

	Median wealth, ages 65–69		Median wealth at death	
Quintile	Data	Model	Data	Model
Q1 (lowest)	130,604	57,374	40,758	210,658
Q2	190,990	122,395	84,479	46,647
Q3	284,409	235,004	66,503	206,836
Q4	426,755	373,178	170,691	288,828
Q5 (highest)	724,077	728,863	127,902	490,183

Notes: Quintiles assigned from mean real earnings at ages 54–56. Data from RAND HRS 1995–2008, survey-weighted, in 2010 dollars. Model moments constructed on the same observed-sample definition using the empirical HRS observer profile.

6.3 Validation

Post-claim work rates. The introduction documents that 40 to 50 percent of ever-claimers in HRS birth cohorts 1935–1948 work at least once between ages 60 and 75 after claiming Social Security. This fact is central to the paper's argument that claiming and retirement are distinct margins. We do not target any post-claim work moment in the estimation. Figure 6 reports the fraction of claimed households who are simultaneously working, by age bin, in both the model and the HRS.

In the HRS, 28.5 percent of respondents receiving Social Security retirement income work at ages 62–64, 29.8 percent at ages 65–69, 20.0 percent at ages 70–74, and 12.1 percent at ages 75–79. The model generates post-claim employment of 40.8, 32.9, 17.4, and 7.1 percent in the same age bins. The model overstates post-claim work at ages 62–64 — where the deterministic claiming schedule delays claiming relative to the data — but captures the broad level and

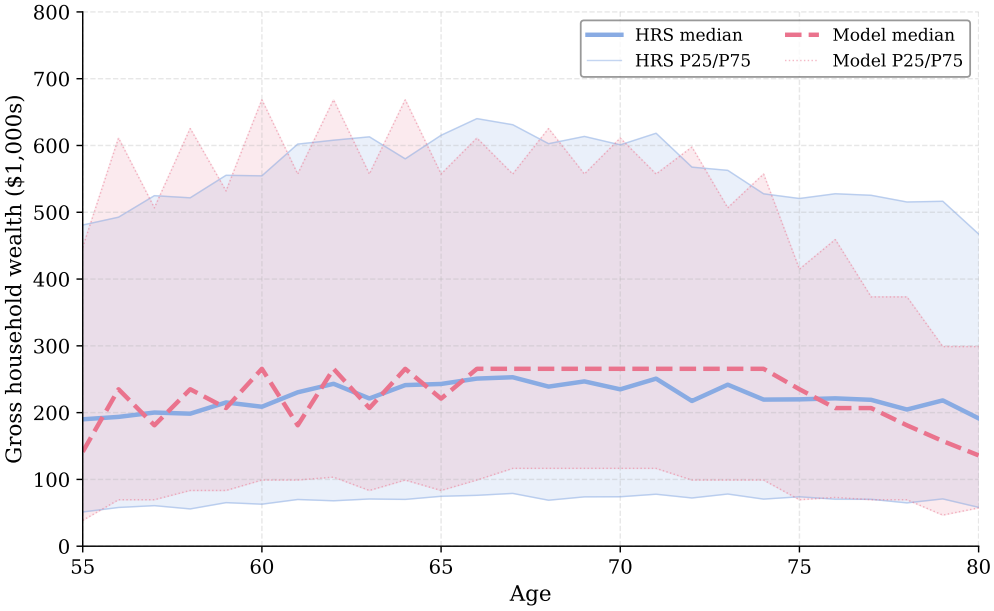

Figure 5: Wealth Percentile Paths by Age: Model versus Data

Notes: Dark shading with dark line: HRS 25th–75th percentile range and median of gross household wealth by age from pooled 1995–2008, survey-weighted. Light shading with red line: simulated model counterparts (1,000 households, weighted by the empirical HRS observer profile). The estimation targets only median wealth at ages 65–69 and the median decumulation to ages 75–79; the full percentile fan at each age is untargeted. All values in 2010 dollars.

declining shape from 65 onward. The model's post-claim work rate at ages 70–74 (17.4%) is close to the HRS counterpart (20.0%), confirming that the estimated preferences sustain work after claiming without targeting this margin.

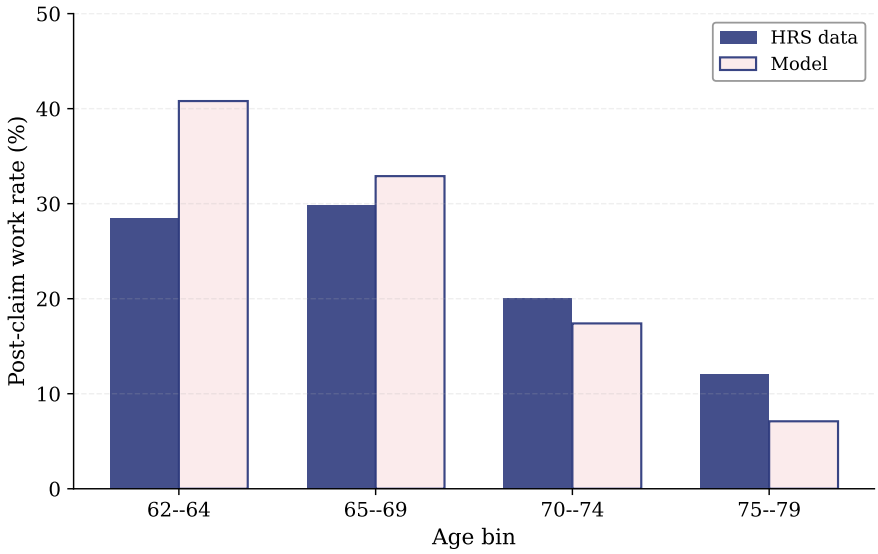

Figure 6: Post-Claim Work Rates by Age: Model versus Data

Notes: Each bar shows the fraction of respondents (HRS) or simulated households (model) who have claimed Social Security and are currently working, among all who have claimed, by age bin. HRS: respondents receiving positive SS retirement income in pooled RAND HRS 1996–2008, survey-weighted. Model: simulated panel (1,000 households) with the empirical observer profile. These moments are not targeted in the SMM estimation.

7 Policy Counterfactuals

8 Robustness and External Validation

9 Conclusion

Social Security solvency risk changes behavior before any reform is enacted. Once claiming and retirement are treated as separate margins, the logic is straightforward. Higher cut risk makes it more attractive to front-load protected benefit payments through early claiming, but it also makes continued work and private saving more valuable as self-insurance against lower future benefits. The same shock can therefore increase claiming without implying earlier retirement.

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A Appendix: Tables and Figures

This appendix provides supplemental material organized by the main-text section it supports. Section A.1 reports the annual solvency-risk index values. Section A.2 describes the numerical solution method. Section A.3 documents the initial-condition construction. Section A.4 gives the age-earnings profile estimation. Section A.5 compares the model's survival inputs with SSA life tables. Section A.6 reports the full target vector and estimation search block. Section A.7 reports the claiming-profile validation.

A.1 Solvency-Risk Index

Table 7 reports the annual values of the standardized solvency-risk index G_t used in the structural estimation. The index is constructed from SSA Trustees Reports, CBO Social Security baselines, and FRED macro data as described in Section 4.1. Values are reported in the standardized form (approximately mean zero and unit standard deviation over 1995–2019) that enters the logistic cut-hazard function $p_j^c = h(G_{c+j}; \alpha_0, \alpha_1)$.

Table 7: Annual Solvency-Risk Index Values

Year	G_t	Context	Year	G_t	Context
1995	−0.35	Baseline period	2011	+0.87	Post-crisis plateau
1996	−0.30		2012	+0.46	
1997	−0.42		2013	+0.59	
1998	−0.53	Late-1990s surpluses	2014	+0.77	
1999	−0.36		2015	+0.64	
2000	−1.57	Peak optimism	2016	+0.65	
2001	−0.78	Dot-com recession	2017	+0.68	
2002	−0.28		2018	+1.23	Depletion horizon shortens
2003	−0.95		2019	+1.90	
2004	−0.50		2020	+4.48	COVID-19 pandemic
2005	−0.83	1943 cohort turns 62	2021	+2.21	
2006	−0.64		2022	+1.41	
2007	−0.47		2023	+2.70	
2008	−0.25	Financial crisis begins	2024	+2.63	
2009	+0.63	Crisis deepens	2025	+2.68	
2010	+0.83	First cash-flow deficit	2026	+3.08	Depletion within a decade

Notes: G_t is the standardized headline solvency-risk index described in Section 4.1. The index combines a stress factor (weight 0.70) and an uncertainty factor (weight 0.30), each extracted via a single-factor Kalman smoother from official SSA, CBO, and macro inputs, and re-standardized against the 1995–2019 baseline. Positive values indicate worse-than-average fiscal conditions. The 1943 birth cohort's claiming window (ages 62–70) spans calendar years 2005–2013, during which G_t ranges from −0.83 to +0.59. Source: authors' construction from SSA Trustees Reports, CBO Social Security baselines, and FRED.

Table 8 documents the individual series that enter each block of the factor model. All series

Table 8: Solvency-Risk Index: Component Series by Block

#	Series	Construction	Source
Panel A Stress block			
1	75-year actuarial balance	Present-value income minus cost as % of taxable payroll; sign flipped so deficit → higher stress	SSA Trustees
2	Open-group unfunded obligation	Dollar magnitude of long-run shortfall (trillions)	SSA Trustees
3	Years to depletion (inverted)	Projected trust-fund exhaustion year minus current year; sign flipped so shorter horizon → higher stress	SSA Trustees
4	Payable-benefits shortfall	1 − ratio of payable to scheduled benefits at depletion (e.g., 0.23 if 77% payable)	SSA Trustees
5	Trust-fund ratio weakness	Minimum projected reserve ratio over next 10 and 25 years; lower ratio → higher stress	SSA Trustees
6	Macro payroll-base stress	Combines year-over-year wage-bill growth decline with unemployment rate level	FRED
Panel B Uncertainty block			
1	Scenario width	Low-cost minus high-cost depletion year from Trustees alternative scenarios	SSA Trustees
2	Stochastic depletion spread	Width of simulated probability distribution over depletion year	SSA Trustees
3	Cross-vintage revision volatility	12-month rolling std. dev. of release-to-release changes in actuarial balance, unfunded obligation, depletion year, and scenario width	SSA Trustees
4	SSA–CBO disagreement	Absolute gap between SSA and CBO projected depletion years	SSA, CBO
5	Macro revision risk	Short-run changes and rolling volatility of the macro payroll-base stress series	FRED
6	Supervised text signal[a]	Sentence-level uncertainty score from Trustees, CBO, and congressional hearing texts	SSA, CBO, GPO

[a] Enters the uncertainty block only if a preregistered audit gate passes on label coverage, inter-annotator agreement, holdout precision/recall, and calibration. See Section 4.1 for details.
Notes: Each series is standardized within series (mean zero, unit variance) before factor extraction. Within each block, a single latent factor is extracted via the Kalman smoother described in equations (11)–(12). The stress factor is sign-identified to the actuarial balance; the uncertainty factor to the scenario width. The headline index is $G_t = 0.70 \times f_t^{\text{stress}} + 0.30 \times f_t^{\text{unc}}$, re-standardized against 1995–2019.

are signed so that higher values indicate a more adverse environment before standardization.

A.2 Numerical Solution

The model is solved by backward induction from terminal age $J = 90$ to entry age $j = 55$. The state space consists of $N_a = 200$ asset grid points on [0, 2,000,000] with curvature parameter 3.0, $N_A = 30$ AIME grid points on [500, 10,000] monthly dollars, two claim states (unclaimed

and claimed, with claim age stored), $N_\theta = 5$ permanent earnings types, $N_H = 3$ health states, $N_\gamma = 3$ persistent consumption-weight types, and, when the solvency-risk block is active, two policy-regime states ($\omega \in \{0, 1\}$).

The work choice is handled by direct enumeration over the two branches $n \in \{0, 1\}$. When claiming is endogenous, the claim choice $\iota \in \{0, 1\}$ is likewise enumerated for eligible households. Conditional on each feasible discrete branch, the continuous savings decision is solved with the endogenous-grid method (EGM). Because the value function is not globally concave in assets, we apply an upper-envelope step on the common asset grid to select the global optimum across branches. The borrowing constraint is handled separately in the constrained region before the branchwise envelope is formed.

To construct model moments, we simulate $N_{\text{sim}} = 50{,}000$ households forward from the age-55 joint type-health distribution and apply the empirical HRS observer profile when aggregating moments. This ensures that the simulated moments match the interview-parity and age-composition rules used on the data side.

A.3 Initial Asset and AIME Supports

The age-55 entry conditions for assets and the model's Social Security state are estimated from the RAND HRS (waves 3–15, ages 54–56, weighted, real 2010 dollars) before the household problem is solved. Each respondent is assigned one fixed permanent-income type θ_i using mean real earnings at ages 54–56. Table 9 reports the per-quintile weighted medians and within-quintile log standard deviations.

For assets, the model's single asset a is identified with non-housing wealth. For the Social Security state, the public HRS does not directly report accrued age-55 AIME. We construct a quintile-level AIME-equivalent proxy from the HRS Prospective Social Security Wealth file, which reports prospective respondent own-benefit wealth at FRA based on linked SSA Summary Earnings Records. We deflate to 2010 dollars, take the weighted median within each earnings quintile, and convert to monthly AIME by dividing by a survival-discounted real annuity factor (evaluated at FRA = 67, $r = 0.029$, gender-averaged 1955-cohort SSA life table; factor ≈ 8.61) and inverting the PIA bend-point formula used by the model ($\kappa_1 = 761$, $\kappa_2 = 4{,}586$).

A.4 Age-Earnings Profiles $\bar{y}_j(\theta, H)$

The model requires an annual earnings profile conditional on the single work state, measured in constant 2010 dollars. We estimate this object from HRS data over ages 55–90 and hold it fixed during structural estimation.

Table 9: Initial Asset Supports and AIME-Equivalent Proxy by Permanent-Income Quintile

Permanent-income quintile	Median liquid assets at 55	log SD (assets)	Median monthly AIME-equiv. proxy	N (assets)
Q1	\$16,972	2.15	\$1,062	3,208
Q2	\$24,536	2.00	\$1,446	3,306
Q3	\$38,004	1.79	\$2,339	3,064
Q4	\$82,940	1.67	\$3,146	2,795
Q5	\$208,769	1.60	\$3,827	2,513

Notes: All values in constant 2010 dollars. "Liquid assets" denotes RAND HRS total non-housing wealth. The SS column reports the quintile-level AIME-equivalent proxy constructed from median respondent own-benefit wealth at FRA. This proxy is a reduced-form entry input, not directly observed accrued age-55 AIME.

Step 1: Permanent-type assignment. We assign permanent type θ_i from pre-retirement HRS earnings. For each respondent with valid positive earnings and positive survey weights at ages 54–56, we compute the person-level mean of real earnings across all valid observations in that age window. Weighted quintile cutoffs define $\theta_i \in \{1, \ldots, 5\}$.

Step 2: Gender-by-type base curves. We define a worker-sample indicator that equals one if the respondent reports working for pay, has positive real annual earnings, positive survey weight, non-missing health, non-missing annual hours, and at least 500 annual hours. For each gender s and permanent-income type θ, we fit a penalized regression for log annual earnings:

$$\log y_{ist} = f^y_{s\theta}(a_{ist}) + \delta^y_{s\theta,t} + g^y_{s\theta}(b_i) + \varepsilon_{ist},$$

where $f^y_{s\theta}(a)$ is a cubic spline in age ($K = 11$ knots at ages 58, 61, 64, 67, 70, 73, 76, 79, 82, 85, 88), $\delta^y_{s\theta,t}$ are period effects, and $g^y_{s\theta}(b_i)$ are cohort effects. The linear age term is omitted to address the age–period–cohort collinearity. Coefficients are estimated by weighted ridge-penalized least squares, with the penalty applied only to the spline terms and selected by leave-one-person-out cross-validation over $\lambda \in \{0.01, 0.1, 0.5, 1, 5, 10, 25\}$.

Step 3: Health deviations. Let $r_{ist} = \log y_{ist} - \widehat{m}_{s\theta}(a_{ist}, t, b_i)$ denote the residual from the fitted base curve. We fit

$$r_{ist} = h^F_{s\theta}(a_{ist})\mathbf{1}\{H_{ist} = 2\} + h^P_{s\theta}(a_{ist})\mathbf{1}\{H_{ist} = 3\} + u_{ist},$$

again by weighted ridge-penalized spline regression, with good health ($H = 1$) as the reference category. We evaluate the base curve at the reference year 2010, add the health deviations, and

impose the monotone ordering

$$\bar{y}_j(s, \theta, \text{Good}) \geq \bar{y}_j(s, \theta, \text{Fair}) \geq \bar{y}_j(s, \theta, \text{Poor})$$

by weighted isotonic projection. To avoid implausible spikes from sparse late-age cells, we also impose weakly decreasing fitted earnings after age 65 within each gender-by-type-health cell.

Step 4: Gender pooling and validation. The model does not carry gender as a state variable. The final object is pooled using age-specific HRS gender shares within each quintile:

$$\bar{y}_j(\theta, H) = \omega^M_{\theta j} \bar{y}_j(M, \theta, H) + \omega^F_{\theta j} \bar{y}_j(F, \theta, H).$$

Figure 7 shows the resulting work-state earnings profiles by permanent-income quintile and health status. Earnings peak in the late 50s to early 60s and decline gradually with age. Within each quintile, workers in good health earn more than workers in fair or poor health, and the health gradient widens at older ages. The level differences across quintiles are large: at age 60, median annual earnings in Q5 are roughly four times those in Q1. Figure 8 validates the pooled age shape against CPS ASEC age-earnings indices for wage and salary workers ages 55–80.

A.5 HRS Survival Calibration

Table 10 compares the annual survival probabilities used in the model with the SSA period life table from the 2014 Trustees Report (calendar year 2010). The SSA column reports the population-average survival probability $1 - q(x)$, averaging male and female death probabilities over each five-year age bin. The HRS columns report the model inputs by income quintile.

We estimate survival and health transitions on the same fixed-θ_i HRS transition sample, using all available waves (waves 3–15, 1996–2020) for the widest possible age support. For each gender and current-health state, we fit a weighted ridge logit for the two-year death probability on a quadratic in age, permanent-income-type indicators, and age-by-type interactions. We convert to annual survival assuming a constant within-interval hazard: $s_j(h, \theta) = \sqrt{1 - q_2(h, \theta, j)}$. Among survivors, we fit a weighted ridge multinomial logit for next-interview health on the same controls. The gender-specific annual objects are pooled using empirical HRS gender shares within each (θ, h) cell.

Three patterns stand out. First, the HRS inputs lie above the SSA population benchmark in every age bin. This is expected because the HRS sample conditions on respondents with positive pre-retirement earnings and on current health, and is therefore positively selected

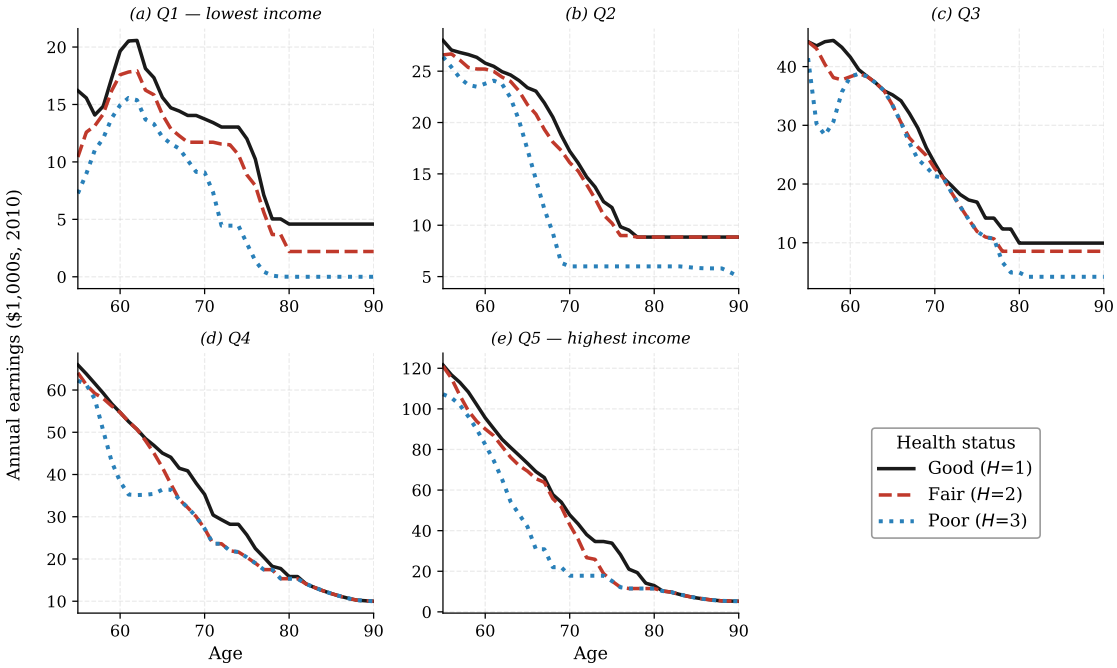

Figure 7: HRS Age-Earnings Profiles for the Work State $\bar{y}_j(\theta, H)$

Notes: Panels (a)–(e) show the HRS-only annual earnings profiles in the model's work state by income quintile and health status. Profiles are estimated from the main worker sample in the HRS (waves 3–15, ages 55–90): working respondents with positive earnings, valid hours, positive survey weights, and annual hours of at least 500, as described in Appendix A.4. Health mapping: Good = HRS excellent/very good; Fair = HRS good/fair; Poor = HRS poor. Appendix Figure 8 adds the pooled CPS age-shape validation for the same HRS-based calibration. All values are in constant 2010 dollars.

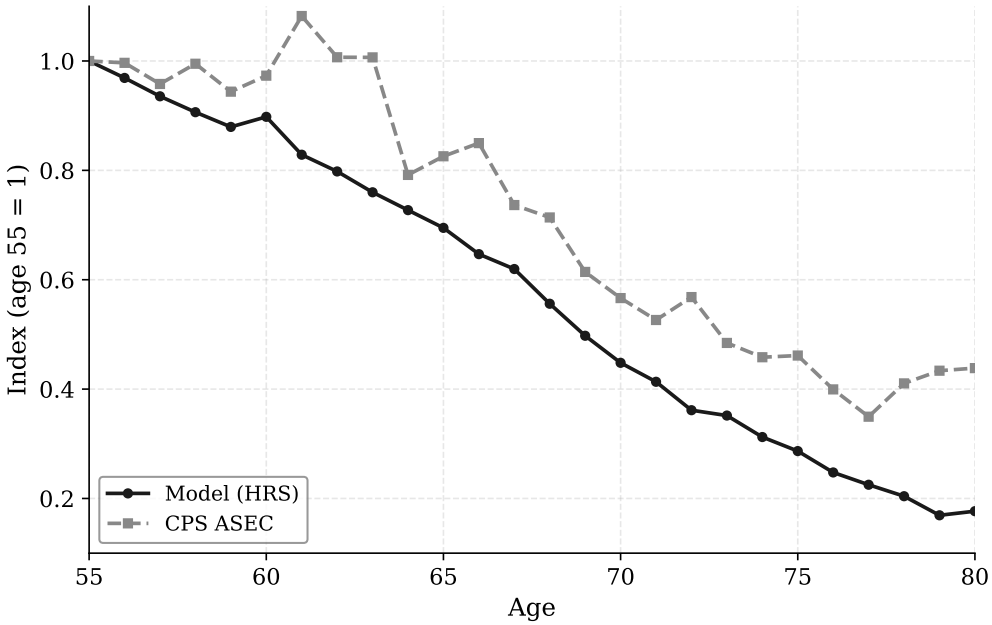

Figure 8: Pooled CPS Validation for the Work-State Earnings Profile $\bar{y}_j(\theta, H)$

Notes: The solid line shows the pooled model-implied age-earnings profile obtained by averaging the final HRS-based $\bar{y}_j(\theta, H)$ profiles across permanent-income types and health states using the empirical HRS type shares and age-specific HRS health shares. The dashed line reports the corresponding pooled CPS ASEC age-earnings index for wage and salary workers ages 55–80 in 2007–2015. Both series are normalized to one at age 55, so the figure validates the age shape rather than the earnings level. All values are in constant 2010 dollars.

Table 10: HRS-Estimated vs. SSA Period Life Table Survival Probabilities

		HRS Estimate				
Age Bin	SSA	Q1	Q2	Q3	Q4	Q5
Panel A: Health-weighted average						
55–59	0.9929	0.9940	0.9946	0.9940	0.9954	0.9962
60–64	0.9898	0.9908	0.9909	0.9913	0.9921	0.9931
65–69	0.9843	0.9862	0.9852	0.9876	0.9882	0.9892
70–74	0.9755	0.9801	0.9776	0.9830	0.9832	0.9847
75–79	0.9605	0.9718	0.9672	0.9770	0.9754	0.9780
80–84	0.9346	0.9595	0.9521	0.9690	0.9629	0.9678
85–90	0.8822	0.9372	0.9252	0.9575	0.9406	0.9503
Panel B: Good health ($H = 1$)						
55–59		0.9970	0.9976	0.9987	0.9989	0.9986
60–64		0.9950	0.9952	0.9970	0.9980	0.9977
65–69		0.9925	0.9916	0.9940	0.9969	0.9966
70–74		0.9898	0.9872	0.9894	0.9955	0.9954
75–79		0.9880	0.9830	0.9834	0.9939	0.9943
80–84		0.9874	0.9802	0.9768	0.9920	0.9936
85–90		0.9874	0.9792	0.9707	0.9900	0.9933
Panel C: Fair health ($H = 2$)						
55–59		0.9916	0.9925	0.9904	0.9949	0.9955
60–64		0.9882	0.9887	0.9887	0.9915	0.9929
65–69		0.9830	0.9825	0.9865	0.9859	0.9885
70–74		0.9746	0.9723	0.9832	0.9765	0.9811
75–79		0.9608	0.9551	0.9781	0.9609	0.9688
80–84		0.9373	0.9255	0.9696	0.9349	0.9485
85–90		0.8908	0.8671	0.9527	0.8867	0.9111
Panel D: Poor health ($H = 3$)						
55–59		0.9723	0.9684	0.9570	0.9305	0.9559
60–64		0.9448	0.9362	0.9145	0.8804	0.9006
65–69		0.9091	0.8957	0.8725	0.8485	0.8491
70–74		0.8697	0.8644	0.8479	0.8396	0.8229
75–79		0.8176	0.8401	0.8253	0.8319	0.7926
80–84		0.7503	0.8101	0.7978	0.8228	0.7499
85–90		0.6662	0.7700	0.7636	0.8112	0.7086

Notes: SSA column: population-average $1 - q(x)$, averaging male and female death probabilities from the 2014 Trustees Report (calendar year 2010). HRS columns: annual survival from gender-specific smooth models estimated on quintile-assigned RAND HRS wave pairs (waves 3–15, 1996–2020, fitted ages 55–79), annualized via $s = \sqrt{1 - q_2}$, and pooled using empirical gender shares. Panel A weights across health states using the distribution from Table 2. Poor-health cells are noisier because approximately 4 percent of the transition sample begins in poor health.

relative to the full population. Second, health is the dominant predictor of survival. At ages 75–79, annual survival in Q1 is 0.988 in good health versus 0.818 in poor health. The health gap is much larger than the within-health income differences. Third, the permanent-income gradient is clear in the health-weighted averages but is not strictly monotone in every poor-health cell at advanced ages, reflecting the small sample size in that stratum.

A.6 Targeted Moments and Estimation Detail

Table 11 reports the estimation search block, including the initialization and admissible bounds for the 11 preference parameters estimated in the second step.

Table 11: Estimation Search Block: Initialization and Bounds

Parameter	Start / bounds	Interpretation
β	0.999; [0.900, 1.000]	Discount factor
σ	2.40; [1.5, 6.0]	Curvature over the consumption-leisure composite
γ	0.431; [0.2, 0.8]	Mean consumption weight in the composite
ψ_0	0.005; [0.003, 0.300]	Baseline work cost
δ	0.05; [0.000, 0.200]	Health work-cost gradient
h_0	0.050; [0.010, 0.500]	Baseline time requirement of work
z	−0.002; [−0.005, 0.100]	Age slope in time requirement
κ_2	0.10; [0.005, 0.150]	Health-related time cost, fair health
κ_3	0.20; [0.010, 0.300]	Health-related time cost, poor health
ϕ_b	0.302; [0.000, 0.990]	Bequest-share shifter
$\underline{c}_b$	\$10,010; [\$10, \$200,000]	Bequest location shifter
$\eta_b = \sigma_b/\sigma$	0.976; [0.30, 0.99]	Bequest-curvature ratio

Notes: The fixed heterogeneity scaffold $(\sigma_\gamma, N_\gamma) = (0.067, 3)$ is outside this search block and is documented in Table 4.

Tables 12 and 13 list the 25-moment target vector, together with bootstrap standard errors and diagonal inverse-variance reference scales.

A.7 Claiming-Profile Validation

Table 14 compares the model's simulated claiming profile with the 2010 HRS cross-sectional claiming distribution at each exact age from 62 through 70. These moments are not targeted in the SMM estimation; they serve as an external check on whether the model's claiming schedule reproduces the empirical timing of benefit receipt.

Table 12: Targeted Moments: Aggregate and Labor

Moment	Target	Bootstrap SE	Diag. scale
Median wealth, ages 65–69	246,160	5,118	3.82e-08
Wealth decumulation, ages 65–69 to 75–79	28,651.4	6,425.96	2.42e-08
Employment rate, ages 55–61	0.675577	0.00468558	4.55e+04
Employment rate, ages 62–64	0.461335	0.00594997	2.82e+04
Employment rate, ages 65–69	0.325434	0.00526247	3.61e+04
Employment rate, ages 70–74	0.199383	0.00448619	4.97e+04
Employment rate, ages 75–79	0.120387	0.00398767	6.29e+04
Employment rate, ages 80–84	0.0605623	0.00343052	8.50e+04
Employment rate, ages 85–89	0.0320009	0.00337670	8.77e+04
Employment rate, ages 90–94	0.0157502	0.00370962	7.27e+04
Nonwork hazard at age 62	0.251265	0.00867168	1.33e+04
Good–fair employment gap, ages 62–70	0.128299	0.00796599	1.58e+04
Good–poor employment gap, ages 62–70	0.375721	0.00925247	1.17e+04
Median wealth at death	116,519	3,487.49	8.22e-08
IQR of wealth at death	305,029	8,476.37	1.39e-08

Notes: All targets from RAND HRS interview years 1995–2008, survey-weighted, in 2010 dollars where applicable. The diagonal scale is the inverse-variance reference; the weighting matrix uses the full regularized bootstrap covariance.

Table 13: Targeted Moments: Permanent-Income Heterogeneity

Moment	Target	Bootstrap SE	Diag. scale
Median wealth, ages 65–69, quintile 1	130,604	19,927.5	2.52e-09
Median wealth, ages 65–69, quintile 2	190,990	17,641.6	3.21e-09
Median wealth, ages 65–69, quintile 3	284,409	25,532.8	1.53e-09
Median wealth, ages 65–69, quintile 4	426,755	38,820.2	6.64e-10
Median wealth, ages 65–69, quintile 5	724,077	56,274.5	3.16e-10
Median wealth at death, quintile 1	40,757.5	23,227.2	1.85e-09
Median wealth at death, quintile 2	84,478.8	11,927.6	7.03e-09
Median wealth at death, quintile 3	66,503.2	25,639.2	1.52e-09
Median wealth at death, quintile 4	170,691	42,223.1	5.61e-10
Median wealth at death, quintile 5	127,902	48,855.4	4.19e-10

Notes: Permanent-income quintiles assigned from mean real earnings at ages 54–56. Same RAND HRS 1995–2008 window as the aggregate moments.

Table 14: Claiming-Profile Validation by Exact Age

Age	HRS target	Model	Error	Bootstrap SE
62	0.278	0.000	-0.278	0.023
63	0.384	0.035	-0.349	0.024
64	0.383	0.048	-0.335	0.030
65	0.547	0.363	-0.184	0.026
66	0.723	0.696	-0.027	0.027
67	0.763	0.881	0.117	0.023
68	0.798	1.000	0.202	0.023
69	0.743	1.000	0.257	0.018
70	0.986	1.000	0.014	0.006

Notes: HRS target: weighted claimed share at each exact age in the 2010 cross section. Model: simulated claimed share among alive households under the deterministic claiming schedule. These are validation objects, not targeted SMM moments.